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New and Improved Probabilistic Star-Galaxy Separation Method for Rubin LSST Catalogs

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ABSTRACT

The morphological classification of resolved and unresolved sources, commonly referred to as "star-galaxy separation", is a fundamental step in the analysis of wide-field imaging surveys such as Rubin Observatory's Legacy Survey of Space and Time (LSST). Star-galaxy separation becomes increasingly challenging at the faint magnitudes probed by LSST because galaxy counts rise rapidly with magnitude while their angular sizes become progressively smaller. LSST photometric catalogs provide several classifiers, including a per-band binary classifier `extendedness` based on the threshold difference between PSF, $m_{\rm PSF}$, and CModel, $m_{\rm CModel}$, magnitudes that was pioneered by SDSS, and a novel method `griz_model_extendedness` that uses best-fit Sersic model parameters and also corrects for noise and size effects. All provided classifiers suffer from several theoretical and practical shortcomings. In this work, we implement the Bayesian star-galaxy classification method of Slater, Ivezić & Lupton (2020, SIL2020), which overcomes these shortcomings. In each band, we model the distribution of sources in the two-dimensional space defined by $\Delta m = m_{\rm PSF} - m_{\rm Sersic}$ and magnitude $m_{\rm Sersic}$. Importantly, CModel magnitudes are replaced by Sérsic magnitudes because the latter result in superior performance. At a fixed magnitude, the distribution is modeled as a mixture of unresolved and resolved components which automatically accounts for the varying star-to-galaxy number ratio and allows us to compute the posterior probability that a source is resolved, as well as sample completeness and contamination for arbitrary selection thresholds. This approach also allows for a statistically sound combination of morphological information from multiple bands. We apply this method to the Rubin Data Preview 2 object catalog and validate the resulting classifications using external HST labels in the COSMOS field with 5σ detection limit for point sources in the r band of 26.4. Using these labels as the ground truth, we quantitatively compare the performance of the Bayesian classifier with that of the default Rubin `extendedness` parameter and find that the former consistently outperforms the latter, with the improvement increasing as additional bands are incorporated. Approximately, the Bayesian classifier achieves the same classification performance for stellar subsamples at SNR limits *more than a magnitude fainter* than the CModel-based `extendedness` classifier, and approaches idealized statistical limit discussed by SIL2020. In addition, it allows users to control sample completeness and/or contamination as a function of magnitude. Depending on scientific applications, this morphological classification can be further improved by incorporating complementary color-based classifiers.

Keywords: Galaxies (573) — Cosmology (343) — Stellar astronomy (1583) — Solar physics (1476)

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1. INTRODUCTION

The morphological classification of resolved and unresolved sources[4], commonly referred to as "star-galaxy separation", is a fundamental step in the analysis of wide-field imaging surveys such as the Rubin's Legacy Survey of Space and Time (LSST). At bright magnitudes this problem is relatively simple, since stars outnumber galaxies, and galaxies that exist are obviously extended on the sky. Star-galaxy separation becomes increasingly harder at faint magnitudes, such as those probed by Rubin's Legacy Survey of Space and Time (LSST; for details see Ž. Ivezić et al. 2019a and references therein) because galaxy counts rapidly increase with magnitude, while their angular sizes become smaller. The selection of clean faint stellar samples with high completeness is especially challenging at high galactic latitudes.

A variety of methods have been proposed in the literature for morphological classification of sources in images. C. T. Slater et al. (2020) (hereafter SIL2020) discussed the statistical foundations of morphological star–galaxy separation and showed that many of the star-galaxy separation metrics in common use today, e.g., by Sloan Digital Sky Survey (SDSS) *photo* pipeline (R. H. Lupton et al. 2002), or SExtractor pipeline (E. Bertin & S. Arnouts 1996), are closely related both to each other, and to the Bayesian model odds ratio, first derived by W. L. Sebok (1979). SIL2020 quantified the scaling of various algorithms with the source noise properties and showed that the Bayesian model odds ratio achieves the best peformance. They also showed how to probabilistically combine multiple measurements, either of the same type (e.g., subsequent exposures), or differing types (e.g., multiple bandpasses), or differing methodologies (e.g., morphological and color-based classification).

Following SDSS, LSST photometric catalogs use a binary classifier `extendedness` in each band, based on the threshold difference between PSF, $m_{\rm PSF}$, and CModel, $m_{\rm CModel}$, magnitudes (described in detail in the next section), $\Delta m = m_{\rm PSF} - m_{\rm CModel}$. For unresolved sources, this Δm is vanishing within noise, with `extendedness=0`, while for resolved objects, selected by $\Delta m > 0.016$ mag, `extendedness=1`. Resolved objects have positive Δm because their PSF flux is smaller than CModel flux due to a narrow weighting PSF profile. For an illustration of Δm distribution in Rubin's Data Preview 1 catalogs, see Figure 28 in Vera C. Rubin Observatory Team et al. (2026) or Figure 3 further below). We also discuss a novel method introduced by Rubin (R. Observatory Science Pipelines Developers et al. 2026), named `griz_model_extendedness`, that uses best-fit Sersic model parameters and additionally corrects Δm information for noise and size effects (more details are available in the next section).

Although `extendedness` was successfuly used in thousands of papers based on SDSS catalogs, this approach is not optimal for LSST. As discussed in SIL2020, when the star-galaxy separation threshold is optimized at the faint, low signal-to-noise ratio (SNR) end of the data, the ability to recognize barely resolved objects at the bright end (such as close binary stars) is not fully exploited. Furthermore, the binary nature of `extendedness` parameter prevents optimal combination of information from different bands, as well as from other classification methods (e.g., based on colors, or other datasets). SIL2020 showed how to turn Δm into a probability distribution, which then allows proper probabilistic combination of multiple bands: with appropriately chosen priors for star-to-galaxy counts ratio, the posterior probability that a source is unresolved (or resolved) can be computed in a statistically sound way using Bayesian framwork.

The main goals of this paper are to implement the steps towards Bayesian star-galaxy separation outlined in SIL2020 using Rubin's Data Preview 2 data and quantify its performance. We introduce our dataset in §2, and in §3 describe star-galaxy separation methodology, including quantification of its performance using HST-based labels. We discuss our results in §4.

2. DATASETS

Here we describe two datasets used in our analysis: Rubin Data Preview 2 catalogs and HST data with star-galaxy labels used for testing methods discussed in §3.

[4] Note that quasars and galaxies can be unresolved and even some stars can be resolved, such as Asymptotic Giant Branch stars. Here we use interchangeably star/galaxy and unresolved/resolved nomenclature when referring to morphological classification.

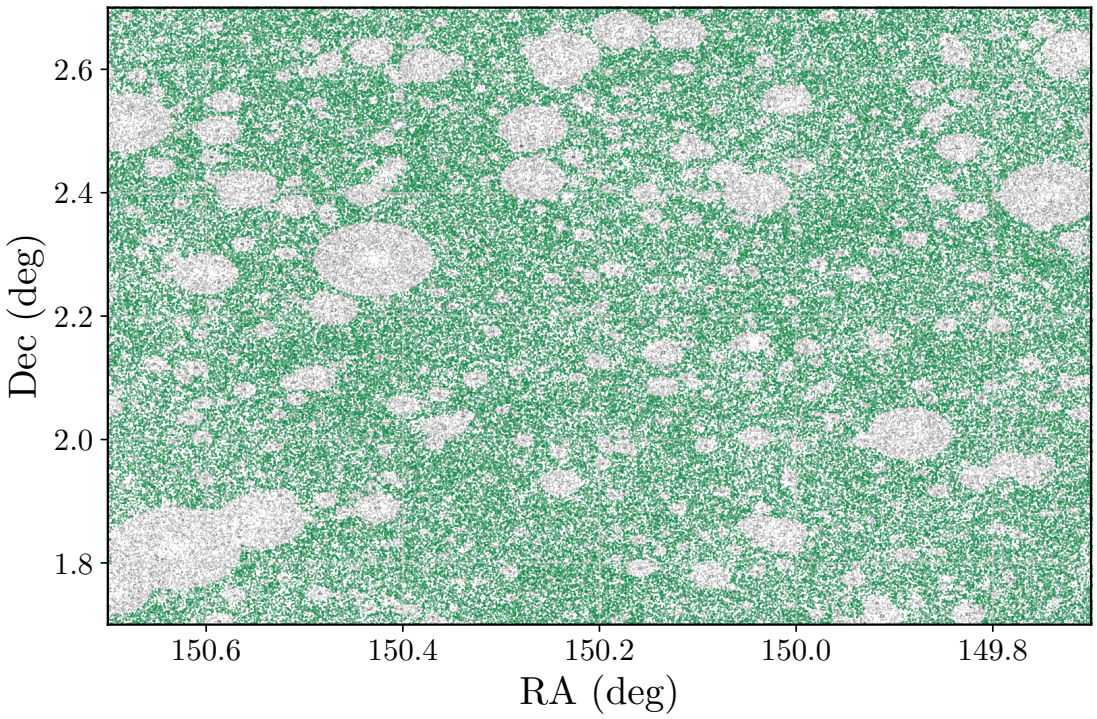


Figure 1. The central DP2-COSMOS region used for the dataset overview. Gray points show Rubin DP2 objects and green points show the subset matched to the HST COSMOS2020/FARMER sample using a 0.5 arcsec matching radius. The white regions in the centers of gray areas correspond to areas where no objects are listed in either catalog.

2.1. *Rubin Data Preview 2*

All data collected with the LSST Camera (Vera C. Rubin Observatory Team 2026) during the commissioning of the Rubin Observatory will be released as Data Preview 2[5] (hereafter DP2). Coadded deep images and resulting catalogs have been released in July 2026 as Early Data Preview 2. These coadded images are based on about 24,000 single-visit images and cover about 3,000 sq. deg. of sky.

In our analysis we focus on the COSMOS field because it is both already well observed by Rubin (to only about 1 magnitude shallower limit than anticipated for 10-year LSST), and space-based HST COSMOS2020/FARMER star and galaxy labels are available (J. R. Weaver et al. 2022). As a control sample for testing code robustness, we also used Rubin data for the Extended Chandra Deep Field South (ECDFS). The Rubin 5σ depth for point sources in the COSMOS field, estimated as the median magnitude of point sources with photometric errors of $\sim$0.2 mag, is (25.66, 26.66, 26.36, 25.91, 24.94, 23.59) in the *ugrizy* bands, respectively.

2.2. *HST Star-galaxy Classification Labels in the COSMOS Field*

For the COSMOS validation sample, we select Rubin DP2 sources in the rectangular region $149.7 < \mathrm{RA} < 150.7$ and $1.7 < \mathrm{Dec} < 2.7$, with $16 \leq r_{\mathrm{CModel}} < 26.5$. Objects listed in the COSMOS2020/FARMER reference catalog are matched to DP2 sources using a 0.5 arcsec matching radius. Within the selected region and magnitude ranges, the DP2 sample contains 269,606 objects, of which 193,586 are matched to the COSMOS2020/FARMER objects and 76,020 are unmatched (a matching fraction of 72%). This somewhat low matching fraction is due to large areas around bright stars that are masked in the COSMOS2020/FARMER catalog (see Figure 1).

[5] See https://dp2.lsst.io/

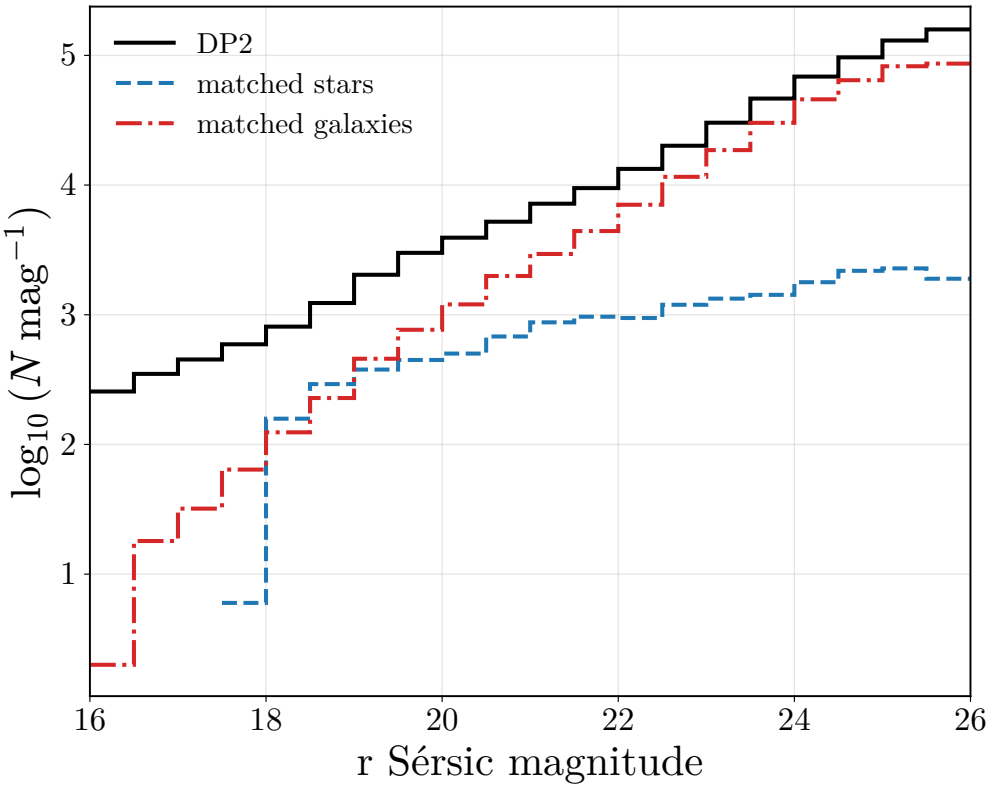


Figure 2. Magnitude distribution of the DP2-COSMOS sample. The black curve shows all selected DP2 sources, while the blue and red curves show matched COSMOS2020/FARMER sources classified as stars and galaxies, respectively. Counts are shown per magnitude, using 0.5 mag wide bins. Note that galaxies outnumber stars at the faint end by more than an order of magnitude.

The matched validation sample contains 8,788 stars and 184,798 galaxies, classified using HST's `truth_binary` = 1 for stars and `truth_binary` = 0 for galaxies. Their number counts as a function of magnitude are shown in Figure 2. The low fraction stars is due to very faint sample limit $r \sim 26$) and high Galactic latitude ($b \sim 50°$).

2.3. *The color-magnitude distributions of sources classified using HST labels*

The external COSMOS2020/FARMER labels provide the reference classification, considered here as ground truth, for examining the distribution of stars and galaxies in various diagrams and for measuring the performance of star-galaxy separation methods. Nevertheless, it is likely that some very faint sources classified by HST as stars might be unresolved galaxies even at HST's resolution. Therefore, the HST labels will help us assess whether a source should be resolved or unresolved in Rubin images, but do not necessarily provide definitive information about astrophysical nature of a source (e.g., to separate unresolved stars and galaxies, one would need spectra or broad-band photometry over a much wider weavelength range, say, in the near-infrared K band). Furthermore, no classification is perfect and thus we first examine the distribution of HST-classified "stars" and "galaxies" in various informative color-color diagrams constructed with Rubin photometry.

The top left panel in Figure 3 shows the distribution of stars and galaxies classified using HST labels in the $\Delta m = m_{\rm PSF} - m_{\rm Sersic}$ vs. $m_{\rm Sersic}$ diagram (justification for replacing CModel magnitude by Sérsic magnitudes in Δm is discussed in §3). As we describe in detail in the next Section, modeling this distribution is a key ingredient of the Bayesian star-galaxy separation method presented here.

The top right panel in Figure 3 shows that faint galaxies ($r > 24$), in addition to outnumbering stars at faint magnitudes, have on average bluer $g - i$ colors (based on CModel magnitudes and corrected for ISM dust reddening using values provided in DP2 catalogs) than stars. The clearly different distributions of stars

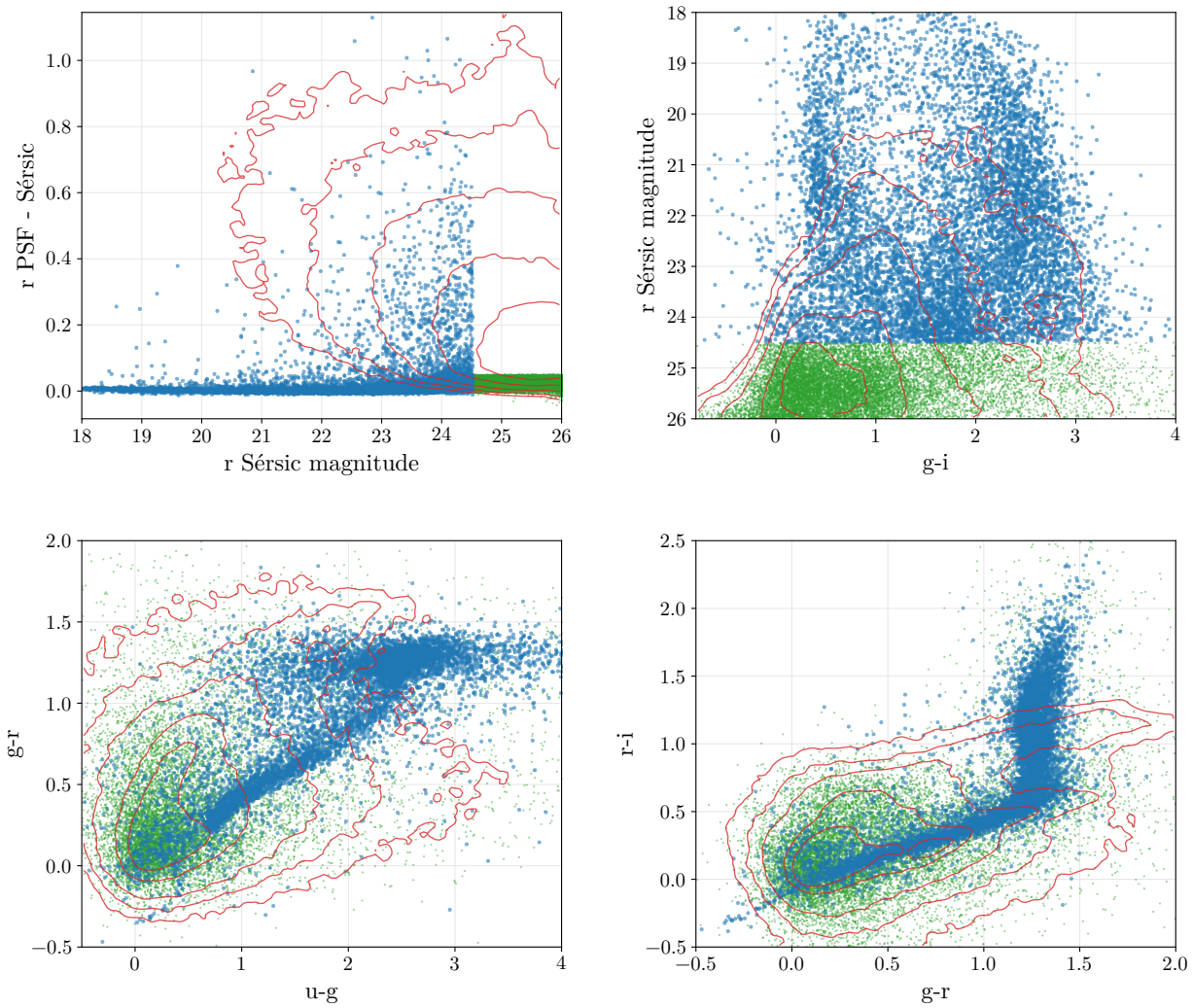


Figure 3. The distribution of stars with r-band Sérsic magnitude < 24.5 (blue dots) and galaxies (red contours), classified using HST labels, in various informative diagrams. Green points show Rubin DP2 sources with r-band Sérsic magnitude $24.5 < r < 26$ and $m_{\rm PSF} - m_{\rm Sérsic} < 0.05$. The CModel-magnitude based colors are corrected for the ISM dust extinction, while the r-band Sérsic magnitude is not. The top left panel shows the $\Delta m = m_{\rm PSF} - m_{\rm Sérsic}$ vs. $m_{\rm Sérsic}$ diagram, where stars form a narrow sequence centered on $\Delta m \approx 0$ and galaxies generally have $\Delta m > 0$. The top right panel shows the r vs. $g-i$ color magnitude diagram, and the two bottom panels show the $u-g$ vs. $g-r$ and $g-r$ vs. $r-i$ diagrams. The main stellar locus, dominated by main sequence stars and red giants, is clearly visible with its blue end redwards from $u-g \sim 0.6$; unresolved sources with $u-g < 0.6$ are expected to be a mixture of low-redshift quasars, white dwarfs and unresolved binaries. In the region with $24 < r < 24.5$ and $0.0 < g-i < 1.0$, galaxies outnumber stars by a factor of 82.

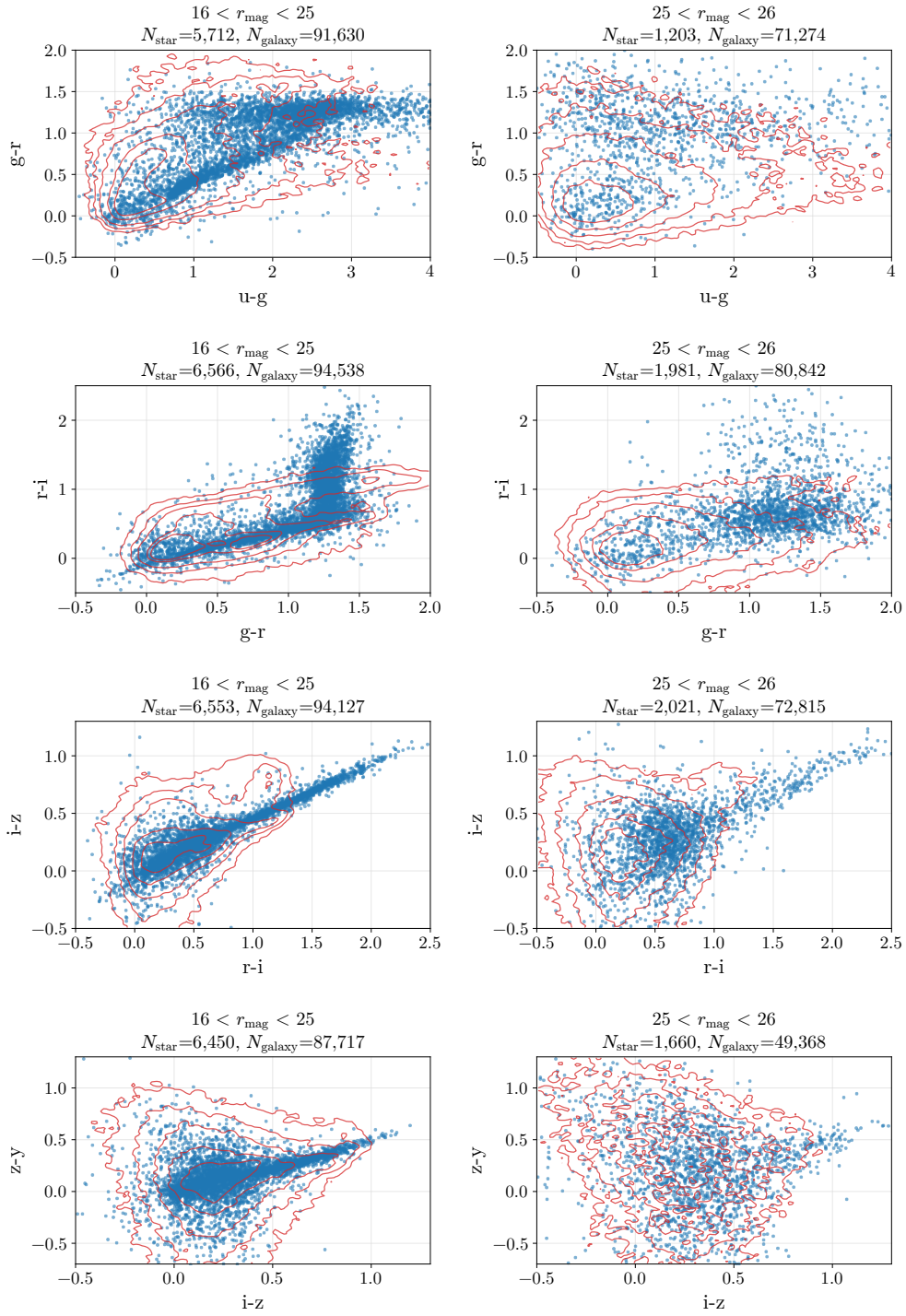


Figure 4. Similar to the bottom two panels in Figure 3, except that here all principal color-color diagrams are shown and the data are split by the r-band Sérsic magnitude to show deterioration of photometry close to the faint magnitude limit (left: $16 < r_{\rm mag} < 25$; right: $25 < r_{\rm mag} < 26$). The 5σ depth for point sources in the r band for this dataset is 26.36.

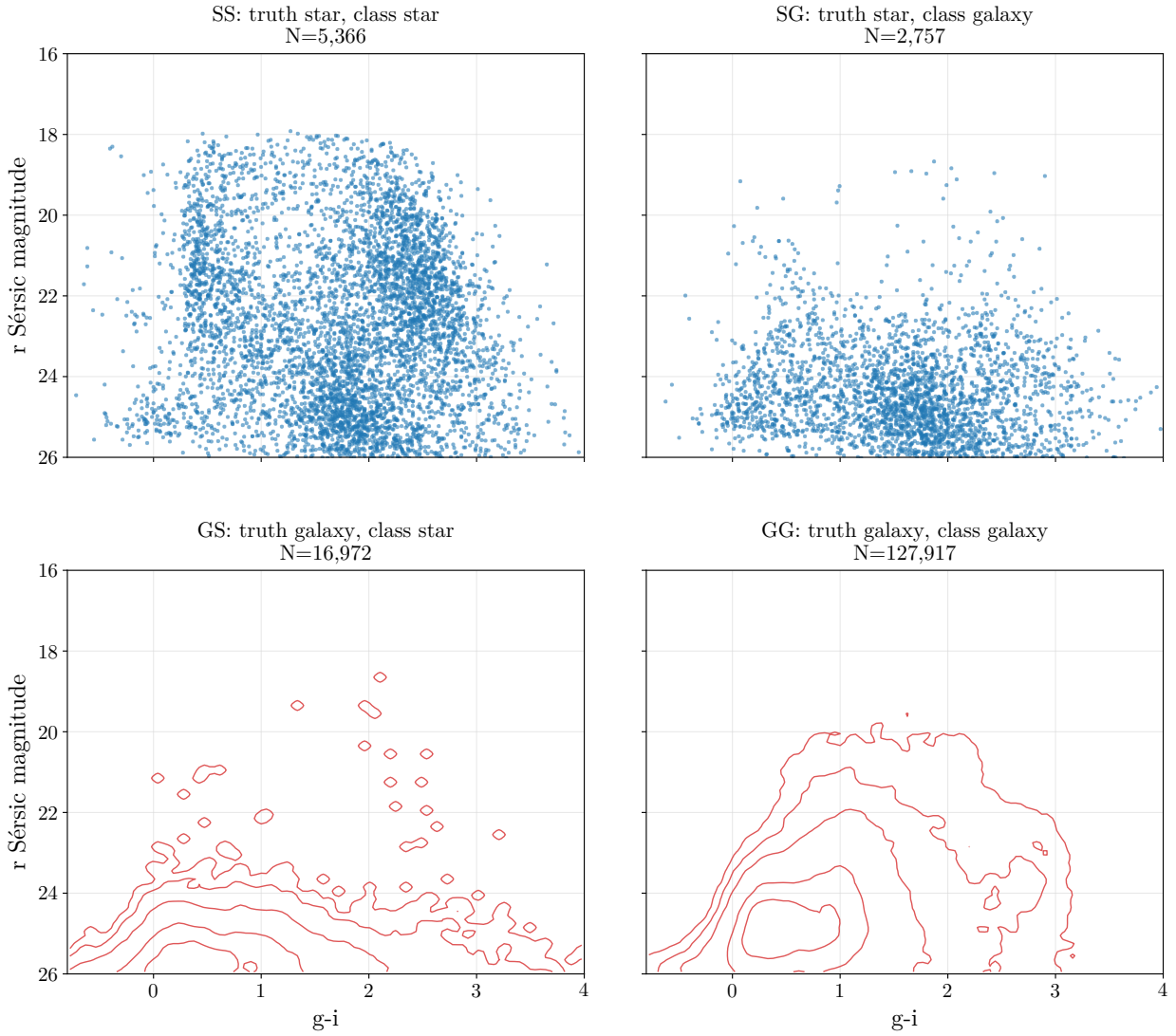


Figure 5. Performance of Rubin's default single-band binary `extendedness` classifier in the r band relative to HST ground-truth labels. The top left and bottom right panels show correctly classified unresolved and resolved sources in the r vs. $g-i$ color-magnitude diagram, with counts shown in the title above each panel. The bottom left and the top right panels show misclassified sources. Note that the stellar sample selected by `extendedness`=0 is severely contaminated by galaxies for $r_{\rm Sérsic} > 24$ (bottom left panel). In the region with $24 < r < 24.5$ and $0.0 < g-i < 1.0$, misclassified galaxies (bottom left panel) outnumber selected stars (top left panel) by a factor of 5.5.

and galaxies in color-color diagrams shown in the bottom two panels in Figure 3 demonstrate that colors can be used to enhance morphological star-galaxy separation. Nevertheless, Figure 4 and the top left panel in Figure 3 show that increased photometric noise close to the faint magnitude limit will cause deteriorated performance for both morphological and color-based star-galaxy separation methods. Extending the start of this deterioration to a fainter magnitude limit is one of the primary goals of this work. It is notable that the top right panel in Figure 4 implies that most unresolved faint blue sources have $u-g$ color more typical of low-redshift quasars ($u-g < 0.6$) than of main sequence stars (for a detailed discussion of the blue star counts see L. Palaversa et al. 2026).

2.4. *The color-magnitude distributions of sources classified using* `extendedness` *parameter*

Difficulties with star-galaxy separation encountered at the faint end are illustrated in Figure 5, which compares subsamples classified using HST ground-truth labels and the default Rubin `extendedness` binary

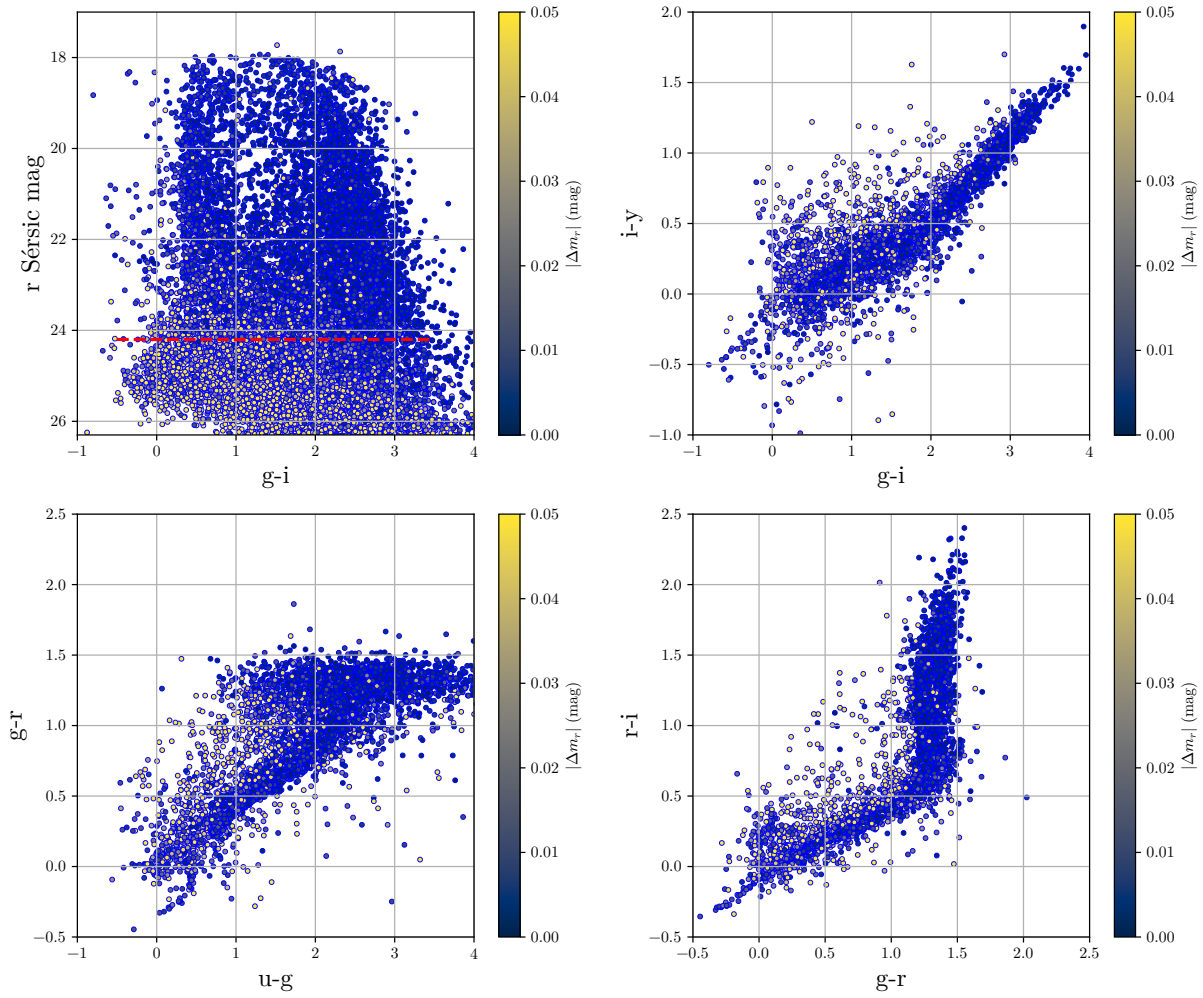


Figure 6. The distribution of objects classified as "stars" by HST in color-magnitude (top left) and color-color diagrams. The sample includes both resolved and unresolved classifications by Rubin. Only objects with $r < 24.2$ are shown in the three color-color diagrams. Symbols are color-coded by the absolute value of Rubin's Δm in the r band. The color-color distribution of objects with $|\Delta m| > 0.05$ mag (about 5% of all objects classified by HST as "stars" and brighter than $r = 23$) is inconsistent with the main stellar locus and similar to that expected for barely resolved or unresolved binary stars.

class in the r band. As evident, a significant fraction of HST sources within about two magnitudes from the Rubin faint limit are misclassified when using default `extendedness` parameter. Misclassifications have a more severe impact on candidate stellar sample: only 26% of "stars" selected by `extendedness`=0 condition are unresolved in HST data. This high level of stellar sample contamination by faint galaxies increases towards the faint end (see the bottom left panel in Figure 5). As the same time, the selection completeness for "stars" (objects unresolved by HST) is only 67%. Due to much higher counts at faint magnitudes and high Galactic latitudes, the purity and completeness for galaxies are much higher: 98% and 89%, respectively. In other words, a simple classifier that declares every object to be a galaxy would formally perform quite well. Therefore, we will focus our analysis in the next Section on the classifier's performance when selecting unresolved objects.

2.5. *Suspect HST Classification Labels*

There is a subset of objects classified as "stars" (unresolved) by HST which appear clearly resolved when using Rubin's default `extendedness` parameter. Figure 6 shows their distribution in color-color diagrams,

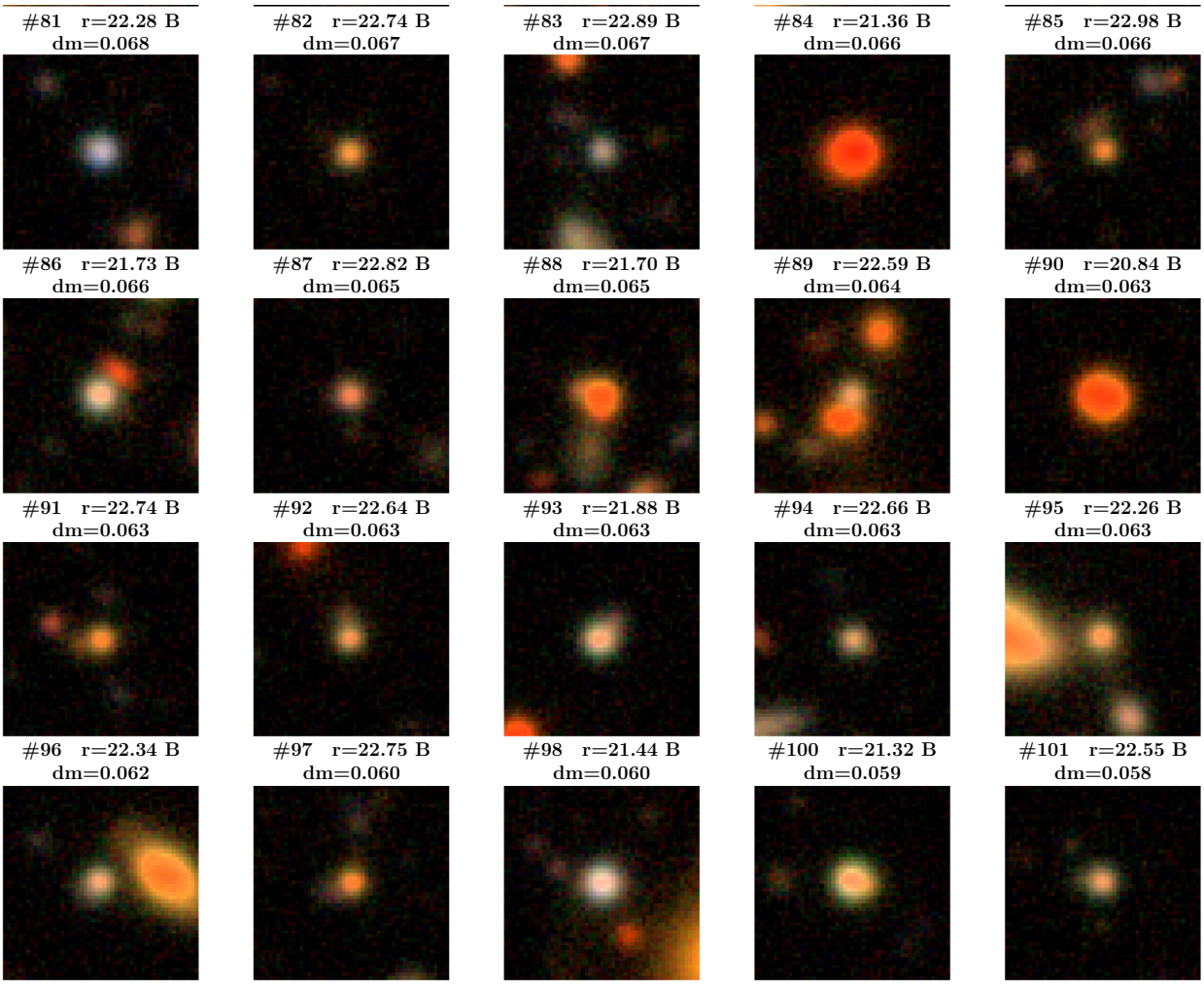


Figure 7. The Rubin *gri* color composite images extracted from "deep_coadd", 10 arcsec by 10 arcsec (50x50 pixels), for a random subsample of 20 (out of 443) HST-classified stars selected using $r_{\rm CModel} < 23$ and $m_{\rm PSF,r} - m_{\rm CModel,r} > 0.05$ mag. About 98.6% of 443 objects are detected as blended objects.

with symbols color-coded by Δm in the r band. As evident, their color-color distribution is inconsistent with main stellar locus and similar to that expected for barely resolved or unresolved binary stars (see Figures 2 and 4 in V. Smolčić et al. 2004.) To test this hypothesis, we visually inspected several hundred images (443 objects with HST label "star", $r < 23$ and $\Delta m > 0.05$ mag; see Figure 7 for an example). Only 1.4% of selected objects are detected as "isolated" (computed using processing flag "detect_isIsolated") and about half seem to have a nearby stellar-like companion. In some cases a bright source or an extended galaxy is nearby, and about one tenth or fewer appear bluish and consistent with quasars. Detailed examination of such cases is beyond the scope of this work.

Because we aim to test statistical performance of our star-galaxy separation method on unambiguous objects, we exclude objects with HST "star" label that both appear resolved in Rubin data and also have Rubin colors inconsistent with the main stellar locus.

We checked whether HST-labeled stars lie near the empirical main stellar locus in Rubin colors quantified using data for stars with $r < 23$ from the COSMOS field. For each source, we calculated its shortest distance to the locus using five dereddened CModel colors: $(u-g, g-r, r-i, i-z, z-y)$. Because the scatter in these colors changes with magnitude, we set a separate distance threshold in each Rubin r-band CModel magnitude bin. Each threshold was set to include at least 80% of HST-labeled stars with measurements in all five colors.

We excluded an HST-labeled star from the performance sample only when it was both beyond the color threshold and classified as resolved by Rubin in the r band (extendedness$_r = 1$). This removed 836 of the 15,760 HST-labeled stars (5.30%), leaving 14,924 (94.70%). In the two magnitude ranges of primary interest for measuring performance, the exclusions were 104 of 1,388 stars (7.49%) at $24 < r < 24.5$ and 116 of 1,700 stars (6.82%) at $24.5 < r < 25$. Stars without the measurements needed to check both criteria were not excluded by this procedure.

3. STAR-GALAXY SEPARATION METHODOLOGY

We first briefly review the Bayesian star-galaxy separation method proposed in SIL2020 and then apply it to single-band Rubin data, followed by its extension to the multi-band case. Our principal measured object property is the difference between PSF, $m_{\rm PSF}$, and model $m_{\rm model}$, magnitudes, $\Delta m = m_{\rm PSF} - m_{\rm model}$. For unresolved sources, this Δm is consistent with zero within the measurement noise, while for resolved objects $\Delta m > 0$ (see the top left panel in Figure 3). A detailed comparison and statistical discussion of this quantity and several other quantities proposed for star-galaxy separation in the literature is available in SIL2020.

3.1. *Which model magnitudes to use: CModel or Sérsic?*

Motivated by successful use of Δm based on CModel magnitudes in thousands to SDSS papers, and the fact that `extendedness` listed in Rubin catalogs is also based on this type of model magnitudes, we initially considered this option, too. A performance comparison with `griz_model_extendedness` parameter revealed that replacing CModel magnitudes by Sérsic magnitudes results in much improved star-galaxy separation (see Figure 8). For pedagogical reasons, all plots presented here are based on Sérsic magnitudes but we have produced both versions of the entire analysis.

Both CModel and Sérsic magnitudes/fluxes are based on fits of the Sérsic brightness profile (e.g., A. W. Graham & S. P. Driver 2005)

$$I(R) = I_e \exp\left\{-b_n\left[\left(\frac{R}{R_e}\right)^{1/n} - 1\right]\right\}. \quad (1)$$

to data. Here $I(R)$ is the surface brightness at projected radius R; R_e is the effective radius enclosing half of the total light; b_n is a normalization constant chosen so that R_e encloses half of the total flux; I_e is the surface brightness at the effective radius, and n is the Sérsic index. The Sérsic index controls the concentration and shape of the profile. An exponential profile with $n = 1$ is often used to describe galaxy disks. A profile with $n = 4$, known as de Vaucouleurs profile, is often used to describe elliptical galaxies and classical bulges.

The main difference between CModel vs. Sérsic fluxes is that the latter are based on best-fit profile where n is a free model parameter, while CModel fluxes are based on a linear combination of two best-fit Sérsic profiles where n is fixed to 1 and 4 ("C" stands for composite[6]), The code takes the best fit exponential and de Vaucouleurs profile fits in a given band and asks for the linear combination of the two that best fits the image. The weighting coefficient (clipped between zero and one) is defined as de Vaucouleurs term, fracDeV, defining the CModel composite flux as

$$F_{CModel} = fracDeV\, F_{deV} + (1 - fracDeV)\, F_{exp}. \quad (2)$$

Figure 9 shows Δm vs. $m_{\rm CModel}$ and Δm vs. $m_{\rm Sersic}$ distributions, separately for unresolved and resolved objects classified using HST labels. Two features are noteworthy. First, HST galaxies dominate the region around Δm ~ 0 at the faint end ($r > 24$) much more strongly in case of CModel magnitudes and are likely explanation for its inferior performance at the faint end. Nevertheless, the distribution of Δm for unresolved sources based on Sérsic magnitudes is much broader and with a median that deviates more from zero than in case of CModel magnitudes. This behavior would result in worse star-galaxy separation performance with Sérsic magnitudes but only when a simple cut on Δm is used for classification. With explicit modeling of these distributions, described below, Δm based on Sérsic magnitudes is clearly superior to its alternative based on CModel magnitudes, especially at the faint end of the sample.

[6] For more details, see https://www.sdss3.org/dr8/algorithms/magnitudes.php

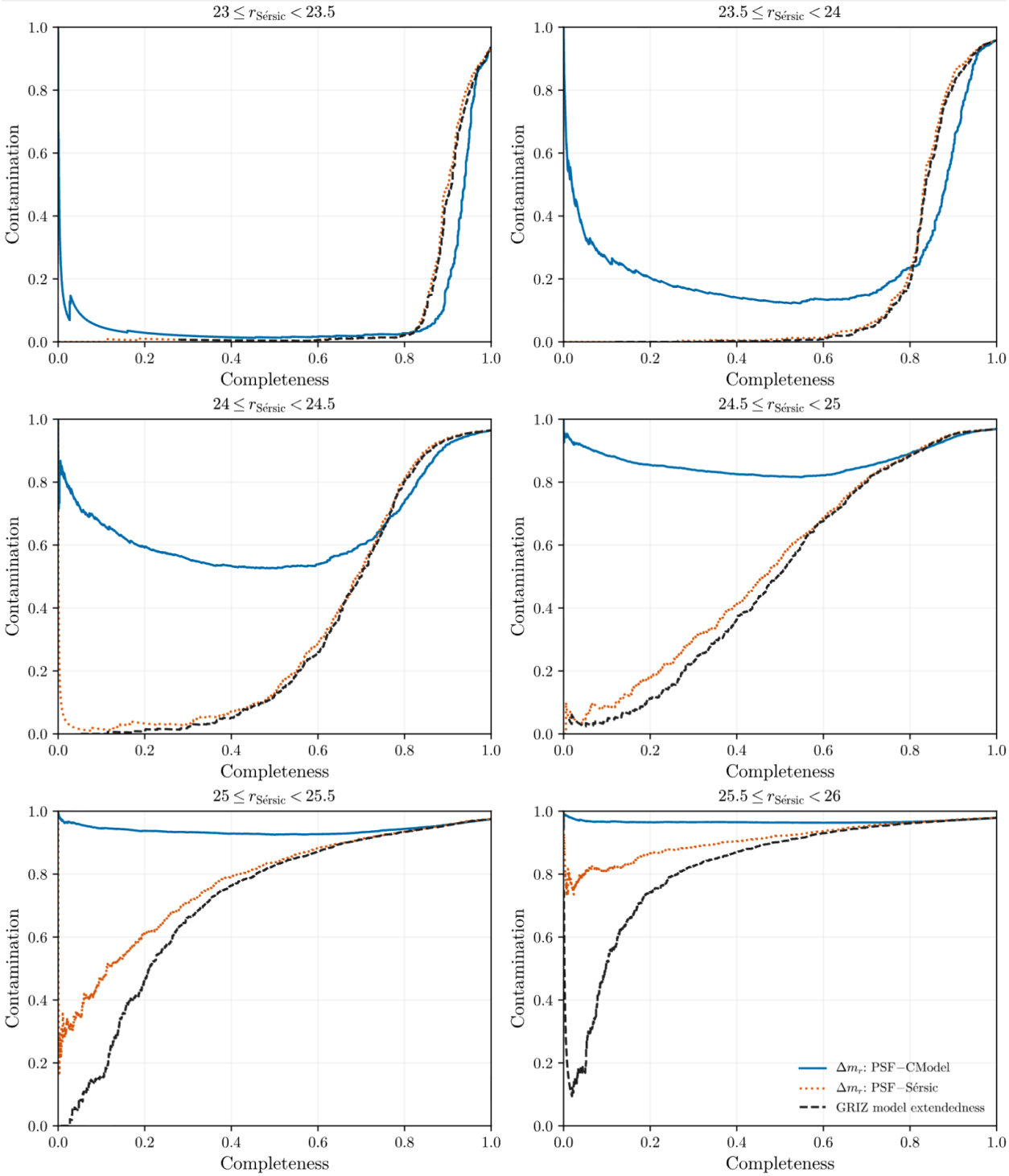


Figure 8. A comparison of contamination (see eq. 5 for definition) vs. completeness (see eq. 6) curves for unresolved sources using two versions of the r-band Δm based on Sersic and CModel fluxes, in different magnitude bins, as marked. Contamination and completeness are estimated using sources with HST labels. Note much lower levels of contamination when Sersic fluxes are used, especially at the faint end.

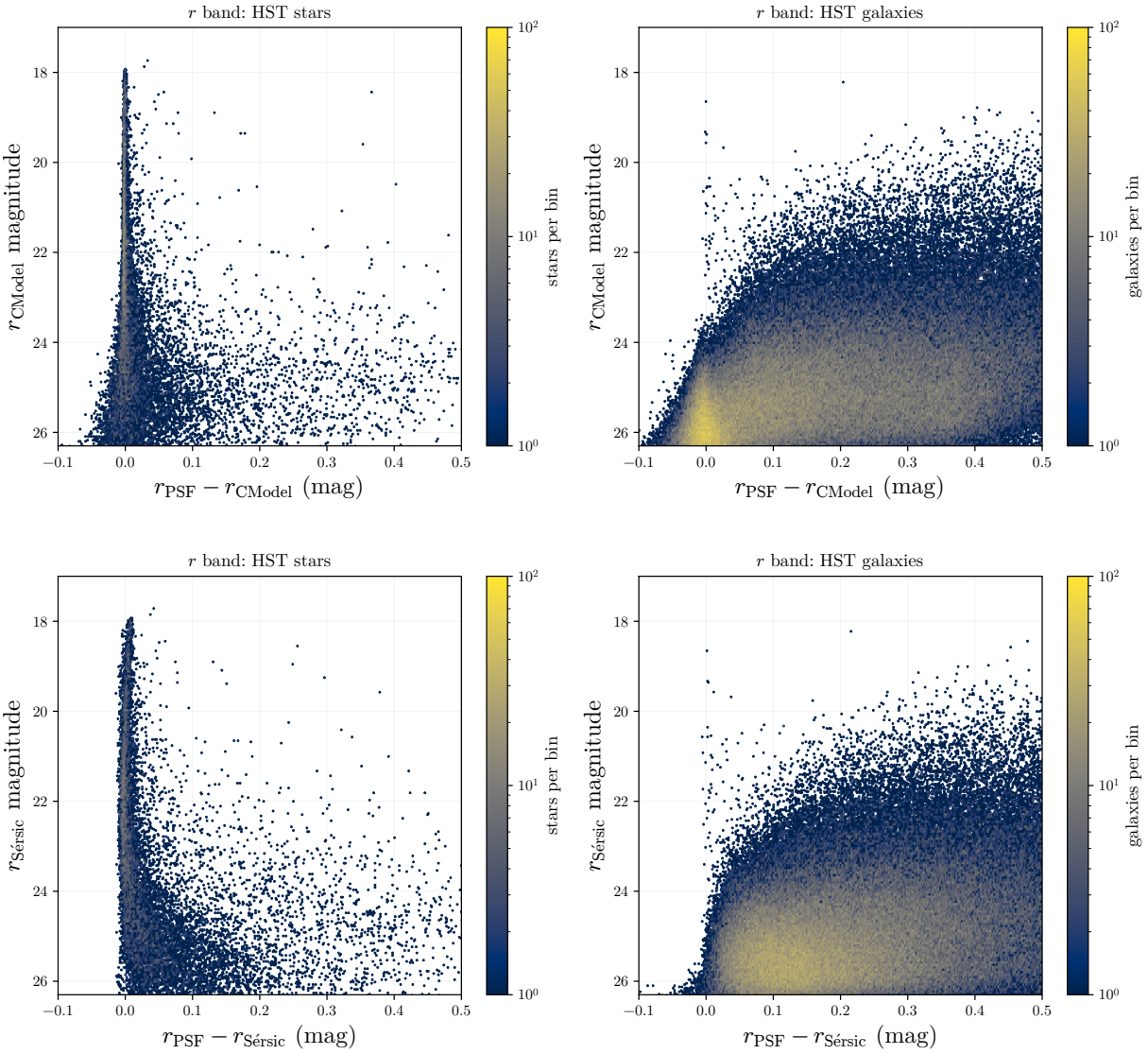


Figure 9. The panels compare the Δm vs. $m_{\rm CModel}$ (top row) and the Δm vs. $m_{\rm Sersic}$ (bottom row) distributions for unresolved (left) and resolved (right) objects classified using HST labels. Note that some objects in the left panels appear clearly resolved by Rubin (Δm > 0.05), even for $r < 24$ (see §2.5). Galaxies dominate the region around Δm ~ 0 at the faint end ($r > 24$) much more strongly in the top right panel than in the bottom right panel. Quantitative modeling of the two-dimensional distributions in the bottom row is at the core of the star-galaxy separation method discussed here.

3.2. *Bayesian Star-Galaxy Separation Method from SIL2020*

We first focus on the single-band case and then extend our discussion to the multi-band case. Using realistic simulations, SIL2020 showed that adopting a fixed, predefined threshold for Δm to select resolved objects is not optimal. The threshold should depend on magnitude because of variations in the object populations and signal-to-noise ratio, as well as on sky location because the galaxy-to-star count ratio varies across the sky. Moreover, for an optimal combination of information from different bands or from other classification methods (e.g., based on colors), the observed Δm distribution needs to be transformed into a proper probability density function.

The subsample of objects with HST labels can be used to quantify the Δm distribution separately for unresolved and resolved objects (see Figure 9). However, such a "training" sample is currently available only for a very small fraction of the LSST footprint (less than 1% of the area). Instead, we use the fact that very large numbers of objects remaining even after binning in both sky location and magnitude can robustly constrain the two components of the Δm distribution when such labels are not available a priori. We only use the subsample of objects with HST labels to independently assess the performance of our method.

3.3. *Mixture Model for Δm vs. $m_{\rm Sersic}$ diagram*

We assume that a sample is selected from a suitably chosen sky region over which the stellar and galaxy populations can be assumed to be statistically uniform. Over angular scales of several degrees, stellar counts vary much more rapidly than galaxy counts (large galaxy clusters are an exception, but not important for the subsequent discussion). Based on the analysis of state-of-the-art Milky Way stellar population models by L. Palaversa et al. (2025), a sky region of several square degrees, corresponding to Rubin tracts[7] is sufficiently small for this purpose (and consistent with our choice of the COSMOS field for quantitative analysis).

Given a "tract" catalog, at a fixed Sérsic magnitude (see below for binning details), the distribution of Δm in a given band (with the band index suppressed for notational simplicity) is modeled as a mixture of an unresolved (star) component and a resolved (galaxy) component,

$$p(\Delta\mathrm{m}\,|\,m) = w_U(m)\, f_{\rm unr}(\Delta\mathrm{m}\,|\,m) + w_R(m)\, f_{\rm res}(\Delta\mathrm{m}\,|\,m), \tag{3}$$

where $w_U + w_R = 1$, and $f_{\rm unr}$ and $f_{\rm res}$ are properly normalized probability density functions. With this decomposition, the probability that an object is unresolved is given by Bayes' theorem as

$$p_S(\Delta\mathrm{m}) \equiv p(\mathrm{unr}\mid \Delta\mathrm{m}) = \frac{w_U(m) f_{\rm unr}(\Delta\mathrm{m}\,|\,m)}{w_U(m) f_{\rm unr}(\Delta\mathrm{m}\,|\,m) + w_R(m) f_{\rm res}(\Delta\mathrm{m}\,|\,m)}. \tag{4}$$

and the probability that it is resolved is $p_G = 1 - p_S$ (note that we use indices S and unr, or G and res interchangeably).

The same model predicts the performance, completeness and contamination, for each class as a function of a hard Δm threshold. If $\Delta\mathrm{m} < \Delta\mathrm{m}_{\rm cut}$ is classified as unresolved (S),

$$\mathrm{Completeness}(S) = F_{\rm unr}(\Delta\mathrm{m}_{\rm cut}) = \int_{-\infty}^{\Delta\mathrm{m}_{\rm cut}} f_{\rm unr}(\Delta\mathrm{m}) d\Delta\mathrm{m}, \tag{5}$$

and

$$\mathrm{Contamination}(S) = \frac{w_R F_{\rm res}(\Delta\mathrm{m}_{\rm cut})}{w_U F_{\rm unr}(\Delta\mathrm{m}_{\rm cut}) + w_R F_{\rm res}(\Delta\mathrm{m}_{\rm cut})}. \tag{6}$$

We test these predictions below using HST-based labels.

Therefore, a good model for $p(\Delta\mathrm{m}\,|\,m)$ (eq. 3) is crucial. Informed by observed Δm distributions (see Figure 9), we model the unresolved component as a weighted sum of a "core" distribution J and a "tail" distribution T.

$$f_{\rm unr}(\Delta\mathrm{m}) = [1 - f(m)]\, J(\Delta m) + f(m)\, T(\Delta m), \tag{7}$$

[7] In the Rubin Science Pipelines, tracts and patches are the two levels of spatial subdivision used for coadded images and object catalogs. They are designed to make processing and storage scalable while preserving continuity across the sky. A tract is a relatively large region of the sky, approximately 1.75° × 1.75° (the exact size depends on the sky map projection). that is processed as a unit. Tracts overlap slightly with neighboring tracts to avoid edge effects. Each tract is subdivided into a regular grid of patches, with each tract in DP2 containing 81 patches. The patch size is about 12 arcmin × 12 arcmin (approximately, one CCD).

where function $f(m)$ quantifies observed strentgth of the tail component. The core component is modeled as

$$J(\Delta m) = \frac{\delta}{\sigma_c(m)\sqrt{2\pi}\sqrt{1+z^2}}\, \mathrm{e}^{-\eta^2/2}, \tag{8}$$

where $z = [\Delta m - \mu_c(m)]/\sigma_c(m)$ and $\eta = \gamma + \delta\ asinh(z)$ ($\delta > 0$). This is a normal density in the transformed coordinate multiplied by $|d\eta/d\Delta m| = \delta/[\sigma_c\sqrt{1+z^2}]$. Increasing $\sigma_c(m)$ generally broadens the curve; γ and δ change its shape.

The tail component is modeled as

$$T(\Delta m) = \frac{1}{hy\sqrt{2\pi}} \exp\left[-\frac{(\ln[y/a(m)])^2}{2h^2}\right] \tag{9}$$

where $y = \Delta m - \mu_c(m)$ and $a(m) = k\sigma_c(m)$, with h and k free model parameters (with γ and δ, the shape of unresolved component is modeled with four free parameters).

Based on data exploration and trial-and-error attempts, we assume the following functions to model variations of normalization functions with magnitude

$$w_U(m) = w_{floor} + \frac{1 - w_{floor}}{1 + \mathrm{e}^{-c-d(m-22)}} \tag{10}$$

and

$$f(m) = \frac{1}{1 + \mathrm{e}^{-f_0 - f_1(m-22)}}. \tag{11}$$

Here w_{floor}, c, d, f_0, f_1 are additional five free model parameters that describe the variation of the tail-to-core and unresolved-to-resolved ratios with magnitude.

The resolved component is modeled as a sum of K=2 skew-normal distributions with weight $w_{R,k}$, location $\mu_{R,k}$, scale $\sigma_{R,k}$, and shape parameter $\alpha_{R,k}$,

$$f_{\rm res}(\Delta {\rm m}) = \sum_{k=1}^{K} w_{R,k}\, \mathcal{SN}(\Delta m|\, \mu_{R,k},\, \sigma_{R,k},\, \alpha_{R,k}), \tag{12}$$

where $\mathcal{N}$ denotes the Gaussian distribution and $\mathcal{SN}$ the skew-normal distribution. Here $\mathcal{SN}(x \mid \mu, \sigma, \alpha) = 2\phi(t)\Phi(\alpha t)/\sigma$, with $t = (x-\mu)/\sigma$, where ϕ and Φ are the standard-normal density and cumulative distribution functions. Therefore, the resolved component shape model includes 7 additional free parameters (note the normalization constraint $\sum_{k=1}^{K} w_{R,k} = 1$).

Details of an iterative fitting algorithm are described in Appendix. In short, in the first step the shape of unresolved component is initially constrained at bright magnitudes ($r < 22.5$) and at $|\Delta m| < 0.05$, while the shape of resolved component is constrained at fainter magnitudes ($23 < 24$) and at $\Delta m > 0.05$. In the final fitting step, the normalization functions $w_U(m)$ and $f(m)$, and only a subset of shape parameters are fit as smooth functions of magnitude (e.g., eqs. 10 and 11 for $w_U(m)$ and $f(m)$, respectively). For more details, see Appendix A.

The quality of fits is illustrated in Figures 10 and 11, which show the mixture model overlaid on the observed Δm histograms in each magnitude bin, and in Figure 12, which shows the 2D data, model, and residual maps in the Δm vs. $m_{\rm CModel}$ plane. The model performs well in the most important interval $\Delta m < 0.15$, where the probability that a source is resolved increase from zero to essentially unity. In particular, the bottom row in Figure 11 shows good agreement of implied model components with subsets of data classified using HST labels. Detailed agreement for $\Delta m > 0.15$ is not as important, as long as the model is properly normalized so that estimates of stellar sample completeness and contamination are reliable (see next section).

The galaxy-to-star count ratio, which serves as Bayesian prior, is

$$N_{GS} \equiv \frac{N_G}{N_S} = \frac{\sum_{k=1}^{K} w_{R,k}}{w_U}. \tag{13}$$

The effectiveness of our fitting procedure is also demonstrated in Figure 13, which compares the implied galaxy-to-star count ratio $N_{\rm gal}/N_{\rm star}$ as a function of magnitude to counts of objects classified using the

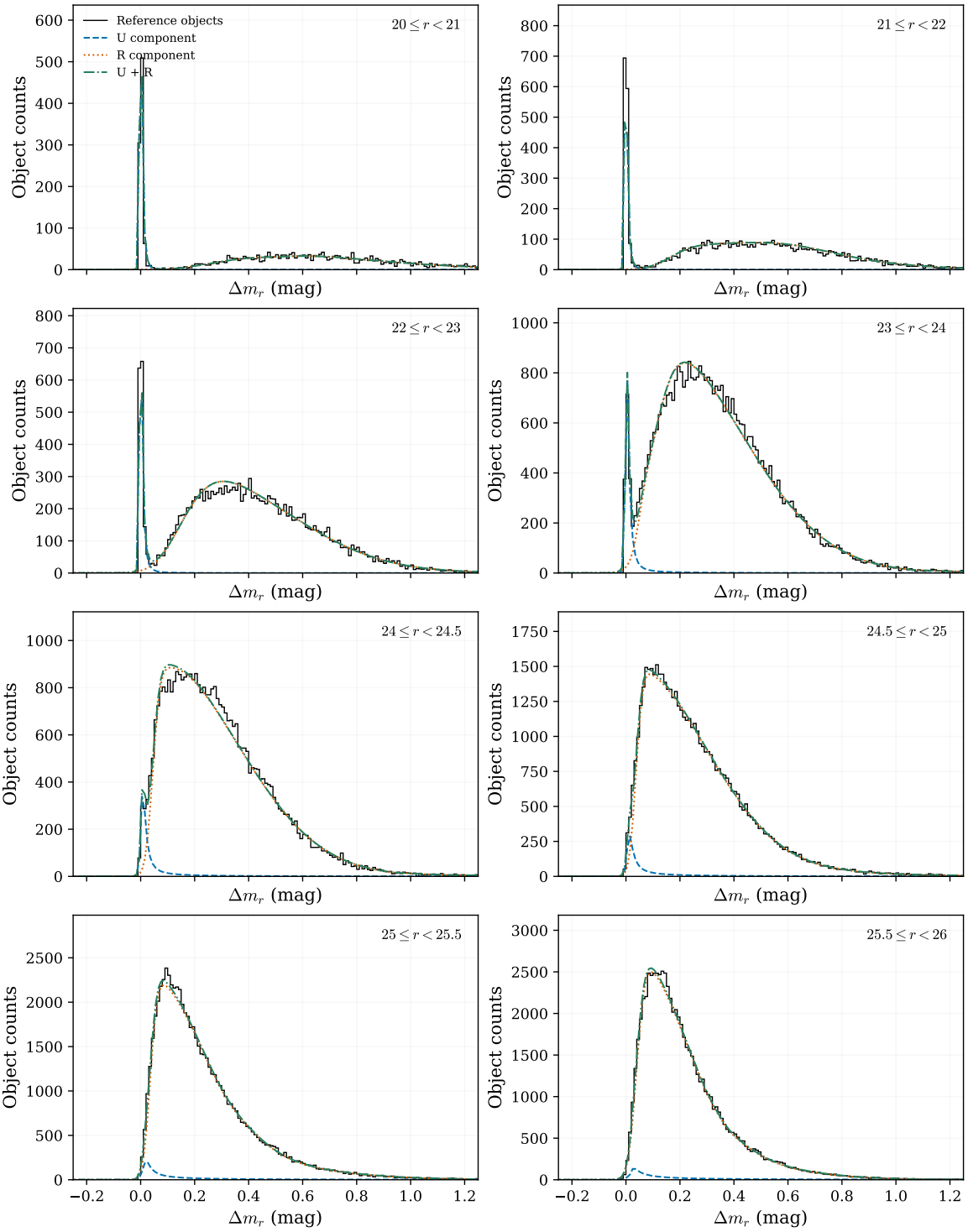


Figure 10. Observed r-band Sérsic-based Δm histograms (black) and the best-fit two-component mixture models (dashed: unresolved component; dotted: resolved component) in magnitude bins, as marked.

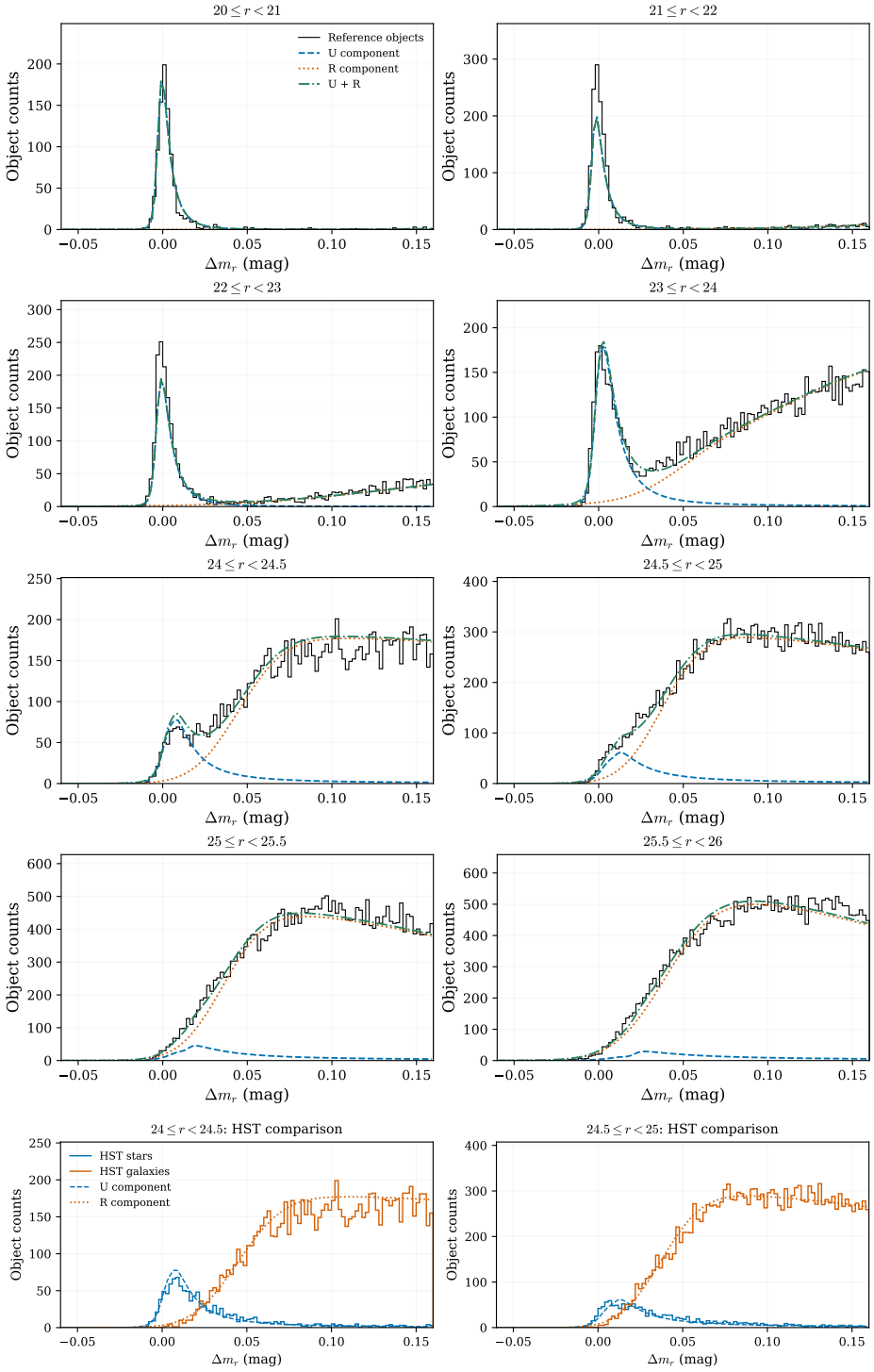


Figure 11. The top four rows show zoomed-in comparison of observed r-band Δm distributions and their best-fit two-component models, in magnitude bins, as marked. Black histograms show the observed counts, while dashed blue, dotted orange, and dot-dashed green curves show the unresolved and resolved components and their sum, respectively. The bottom row compares the two best-fit model components (dashed and dotted smooth lines) to observed distributions for two subsets of source classified using HST labels, in the $24 \leq r \leq 24.5$ and $24.5 \leq r \leq 25$ magnitude bins. Note the excellent agreement between observed and modeled distributions.

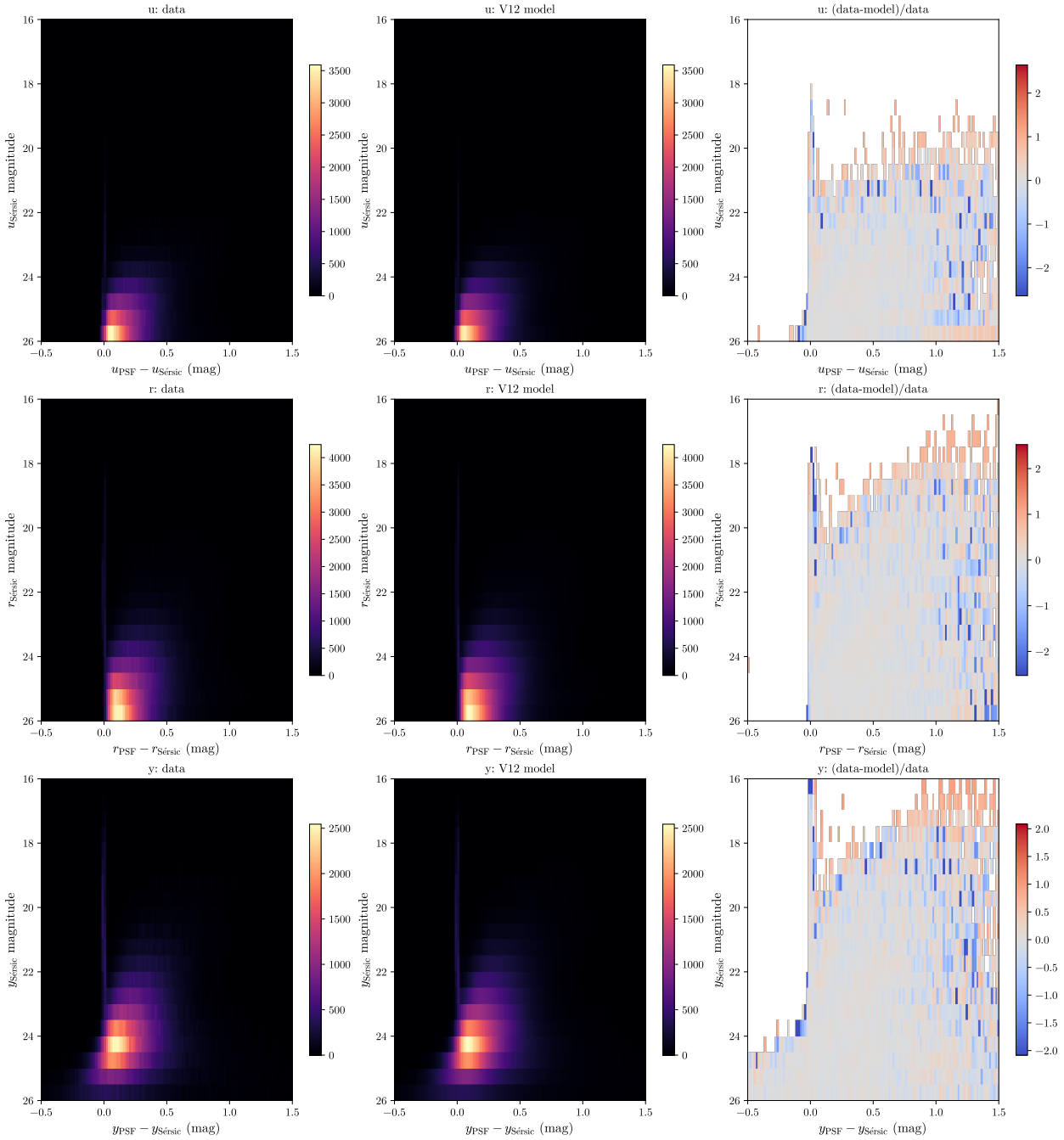


Figure 12. Two-dimensional comparison of the observed (left), modeled (center), and fractional residual (right) source counts in the Δm vs. $m_{\rm Sersic}$ plane, for the u, r, and y bands from top to bottom. Residuals show $(N_{\rm data} - N_{\rm model})/N_{\rm data}$. Results are similar other bands.

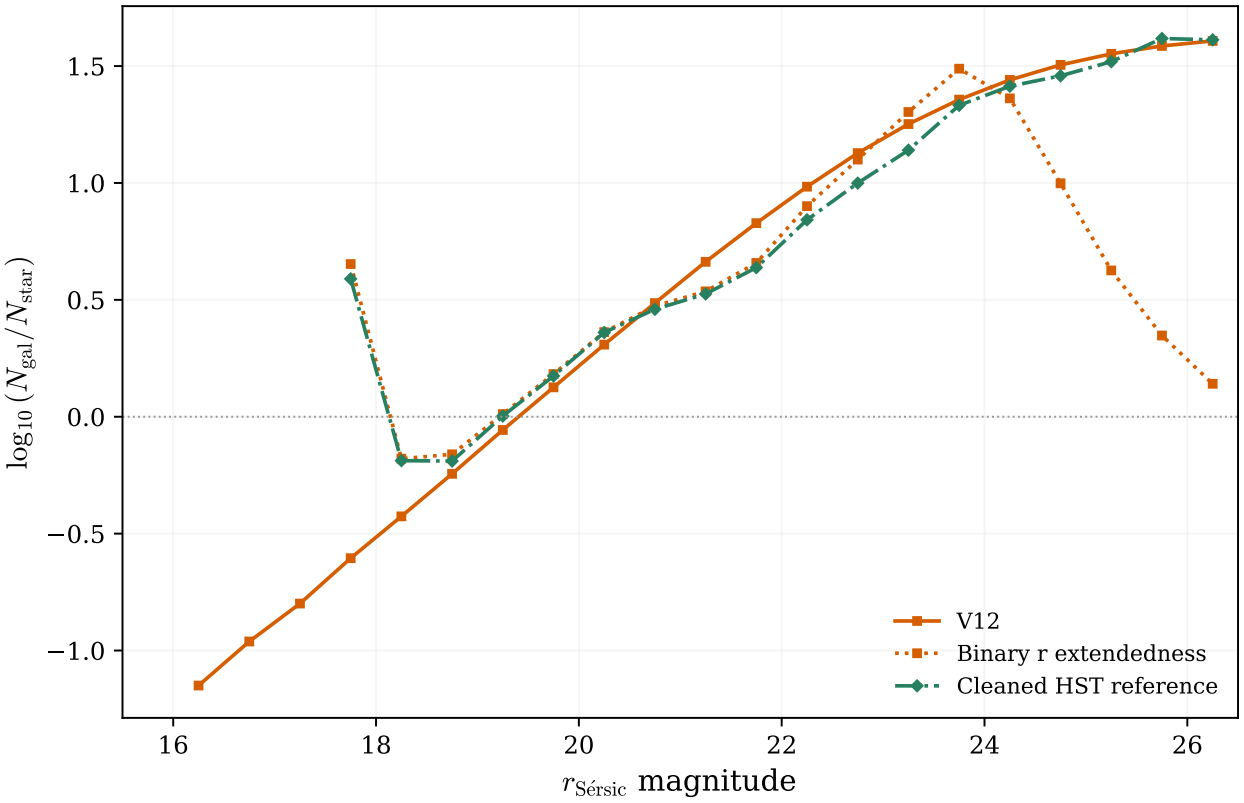


Figure 13. A comparison of model-predicted (solid red line) and HST-observed (green dot-dashed line) galaxy–to-star count ratio as a function of the r-band Sersic magnitude. The counts of sources classified using the r band `extendedness` (dotted line) shows a spurious turnover at $r > 24$ due to a large number of galaxies misclassified as stars.

COSMOS2020 external truth labels (note that in this sample galaxies outnumber stars at the faint end by more than a factor of 30). When binary `extendedness` parameter is used for classification, a spurious turnover is seen at faint magnitudes.

The expression in eq. 4 can be recast as

$$p_S(\Delta\mathrm{m}) = \frac{1}{1 + N_{GS}\, BF}, \tag{14}$$

where the Bayes factor (the likelihood ratio for the galaxy and star hypotheses) is

$$BF = \frac{f_{\rm res}}{f_{\rm unr}}. \tag{15}$$

With $N_{GS} \sim 30$ (see Figure 13), it is sufficient to have $BF \sim 0.033$ for p_S to drop below 0.5 (because the sample is highly imbalanced). This decomposition in terms of prior and Bayes factor is important later when combining multiple sources of classification information (with the overall Bayes factor equal to the product of individual Bayes factors).

3.4. *The Classification Performance of the Single-band Δm Mixture Model*

Given the best-fit model for the data distribution in the Δm vs. $m_{\rm Sersic}$ plane, each object is assigned a probability that it is unresolved, p_S (or resolved, with $p_G = 1 - p_S$). Given a chosen threshold for p_S, the sample can classified into two classes, unresolved and resolved. Importantly, the best-fit model also predicts the class completeness and contamination for the chosen p_S threshold.

Here we use HST labels to evaluate completeness and contamination for the stellar sample and compare them with the mixture-model predictions. We focus on the stellar subsample and the magnitude range,

$22 < r < 26$. Brighter than this range, population mixing is much smaller and measurements more accurate, leading to more accurate star-galaxy separation, whereas at fainter magnitudes the performance deteriorates rapidly (especially for the stellar subsample).

Using the mixture model from eq. 3, we compare completeness and contamination for the stellar subsample predicted by eqs. 5 and 6 with direct counts from the cleaned HST sample in Figure 14. As the figure shows, the agreement betweem the model and data is remarkable, given that HST labels were not used to constrain the model.

Depending on particular science case, a user can control completeness or contamination by choosing p_S threshold. This optimization is discussed in more detail using better-performing multiband classification parameter in §3.8.

3.5. *The Multi-band Case*

We expect classification to improve when Δm measurements from multiple bands are used. When combining information from different bands, a different approach is required than when combining multiple measurements of the same type from a single band. The main reason is that it is sufficient for a source to be reliably identified as resolved in just one band, irrespective of in how many other bands it was unresolved. Therefore, combining information from different bands is a nontrivial statistical problem that is made more complex by astrophysical considerations. For example, galaxy morphology is wavelength dependent: different bands emphasize different stellar populations, dust distribution, star-forming regions, bulges, and disks, and thus a galaxy can show strong resolved structure in one band while being close to a PSF in another. A decisive measurement in one band should therefore not necessarily be diluted by several bands with weak or absent evidence. Consequently, combining measurements from multiple bands is not as straightforward as combining multiple measurements of Bayes Factors (see eq. 15) from individual bands using eq. 14 (see Section 5 in SIL2020 for details).

An optimal Bayesian approach would be to generalize eq. 3 to a vector form and model the true object class as an underlying latent variable. The multi-band measurements are correlated projections of this latent structure rather than independent measurements of one scalar size. We explored a Gaussian copula construction for combining single-band probabilities, but it did not provide a satisfactory description of the desired behavior. We therefore adopt a different approach: first combine the measured Δm values across bands into a single statistic and then derive classification criteria from its distribution, analogously to the single-band case.

Motivated by the fact that the multi-band Δm distribution for the unresolved component can be described as a multivariate nearly-Gaussian with diagonal covariance matrix, we define a standardized variable in each band

$$D_i = \frac{\Delta \mathrm{m}_i - \mu_c(m_i)}{\sigma_c(m_i)}, \tag{16}$$

where i is the band index and μ_c and σ_c come from the best-fit function for unresolved component ($\mu_c \sim 0$). The utilized bands may be all of *ugrizy* or any subset of N usable (depending on data depth and spectral energy distribution of objects of interest) bands, B. We then define a radial metric

$$D = \left(\sum_B D_i^2 \right)^{1/2}. \tag{17}$$

When $N = 1$, $D \sim |\Delta \mathrm{m}/\sigma_c|$, and it measures the distance from the origin in units of σ_c. For the unresolved component, D follows a χ-distribution with one degree of freedom, χ_1 (which is also the half-normal distribution). If μ_U cannot be ignored, D follows a noncentral χ distribution. When $N > 1$, for the unresolved component $D \sim \chi_N$, with an expectation value approximately equal to $\langle D \rangle = \sqrt{N - 1/2}$. Depending on the desired completeness for the unresolved component, a hard D threshold of several times $\sqrt{N}$ can be used for sample classification (this is a standard case of hypothesis testing in frequentist statistics). However, without a quantitative description of the resolved component as well, the resulting sample contamination cannot be estimated.

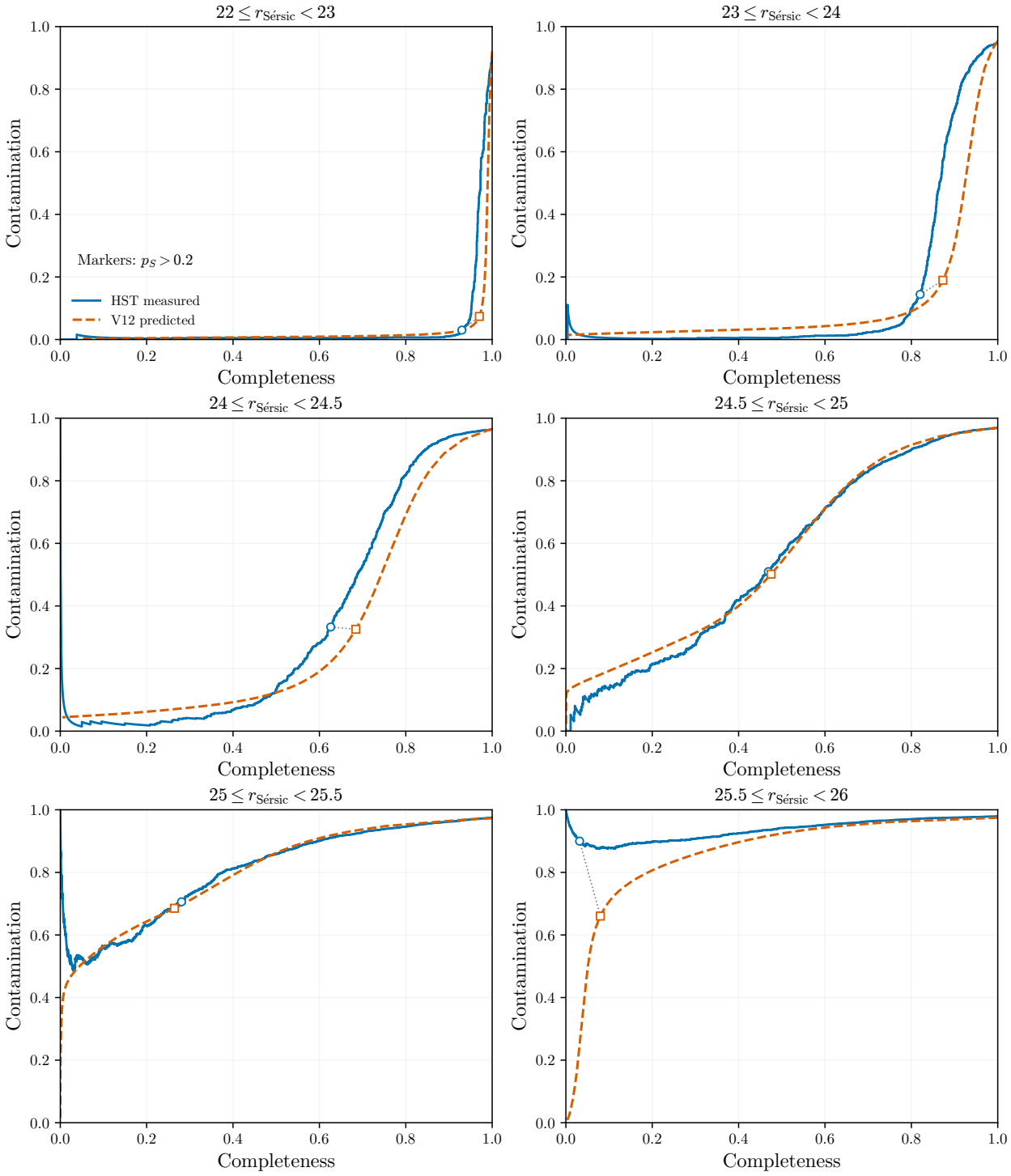


Figure 14. Model-predicted (red dashed lines) and HST-observed (solid blue lines) r-band completeness–contamination relations in different magnitude bins, as marked. Observed curves are based on direct HST counts of unresolved and resolved sources. The model-predicted curves are based on integrals given by eqs. 5 and 6. Open circles and squares mark the HST and model results for $p_S > 0.2$, respectively; dotted segments connect results at the same threshold. Selected samples are highly imbalanced and dominated by galaxies.

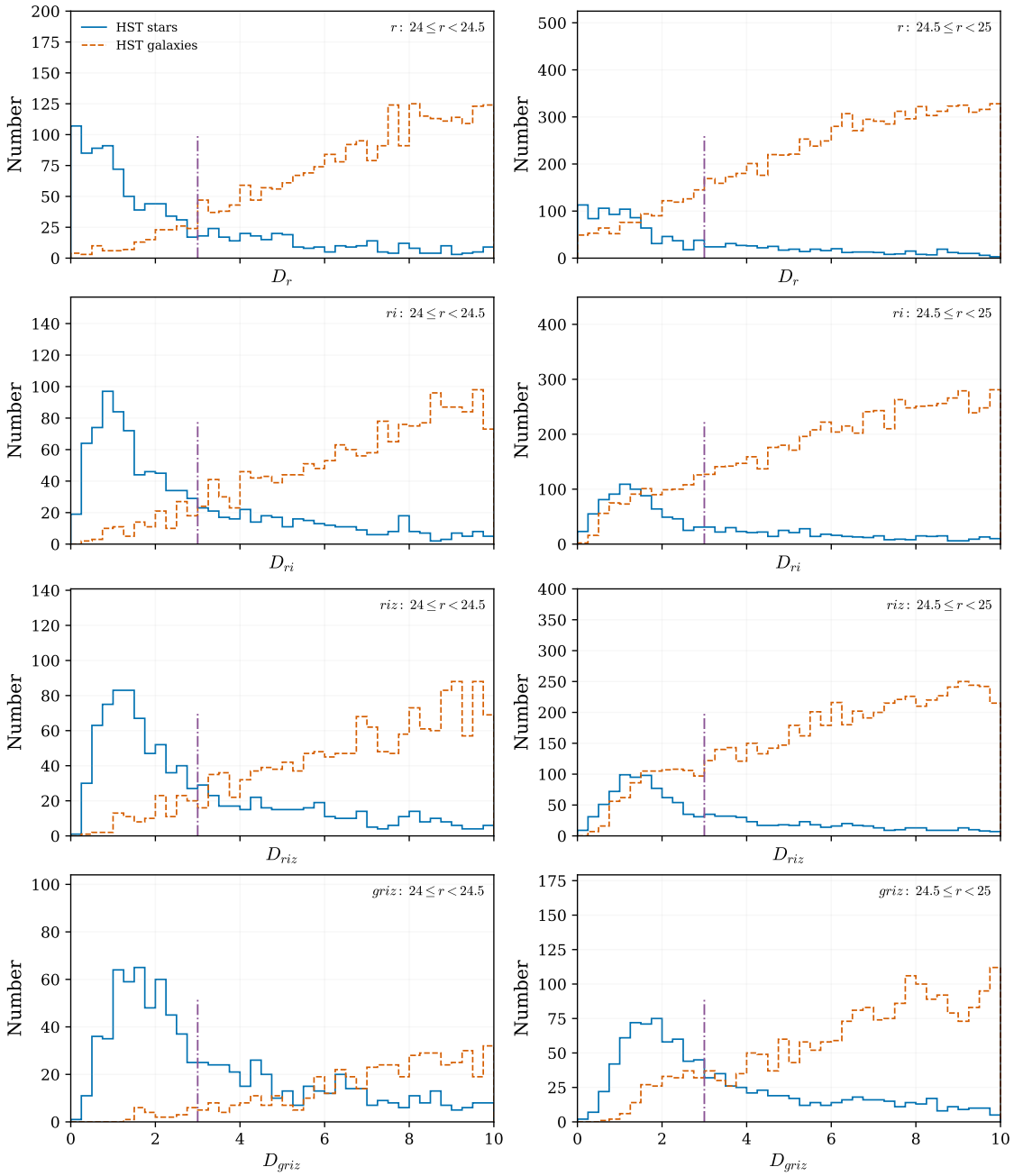


Figure 15. Observed D distributions of HST stars (blue) and HST galaxies (red). Rows show r, ri, riz, and $griz$ band combinations of the D metric; columns show $24 \leq r < 24.5$ (left) and $24.5 \leq r < 25$ (right). Histograms show object counts in bins of width $\Delta D = 0.25$, over $0 \leq D \leq 10$ (distributions for HST galaxies extend to much larger D values of $\sim$100). Half-height vertical lines mark the theoretical χ_N 95% thresholds: $D = 1.960$, 2.448, 2.795, and 3.080 for the four rows, respectively.

3.6. *Two-component Mixture Model for D Distributions*

Figure 15 shows D distributions for HST stars and HST galaxies, and their dependence on the r-band magnitude and band combination. The behavior of D distributions for HST stars is approximately consistent with expectation ($D \sim \chi_N$). We purse fits to the overall D distributions (that is, without using HST labels to define classes) using an analogous two-component mixture model and the same fitting algorithm as we used for Δm distributions.

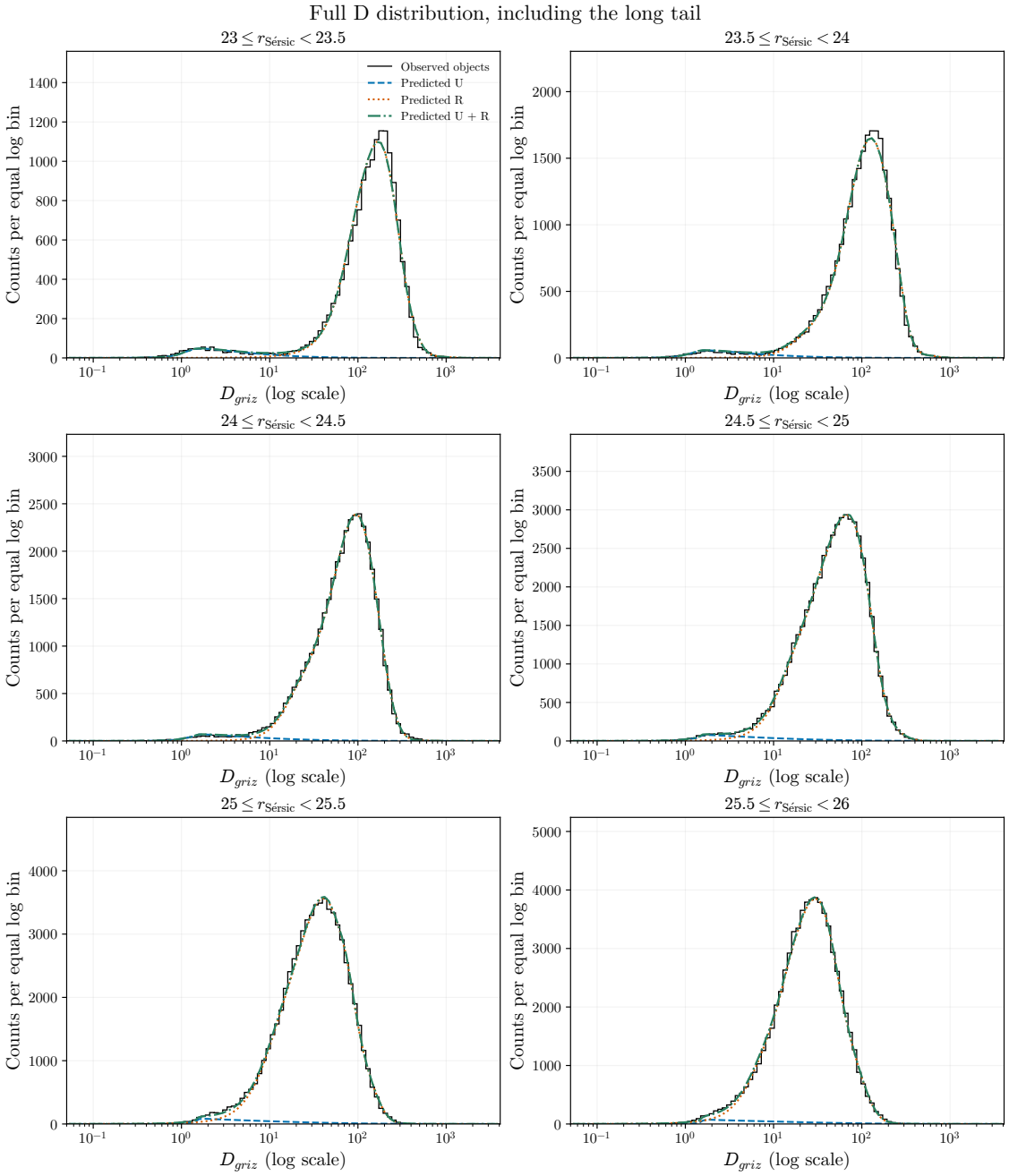


Figure 16. The best-fit two-component models (smooth lines) to observed (histograms) D_{griz} distributions in different magnitude bins, as marked. Note that x axis is shown on logarithmic scale. The large peak corresponds to resolved component; unresolved component peaks at $D_{griz} \sim 2$.

The model performs well, as illustrated in Figures 16 and 17. In particular, the middle column in Figure 17 shows good agreement of implied model components with subsets of data classified using HST labels.

3.7. *The Performance of the D Metric-based Classification*

Here we again use HST labels to evaluate completeness and contamination for the stellar multi-band D-selected samples and compare them to the single-band performance. As before, we focus on the "transition" magnitude range, $23 \leq r < 26$, with six 0.5 mag wide magnitude bins.

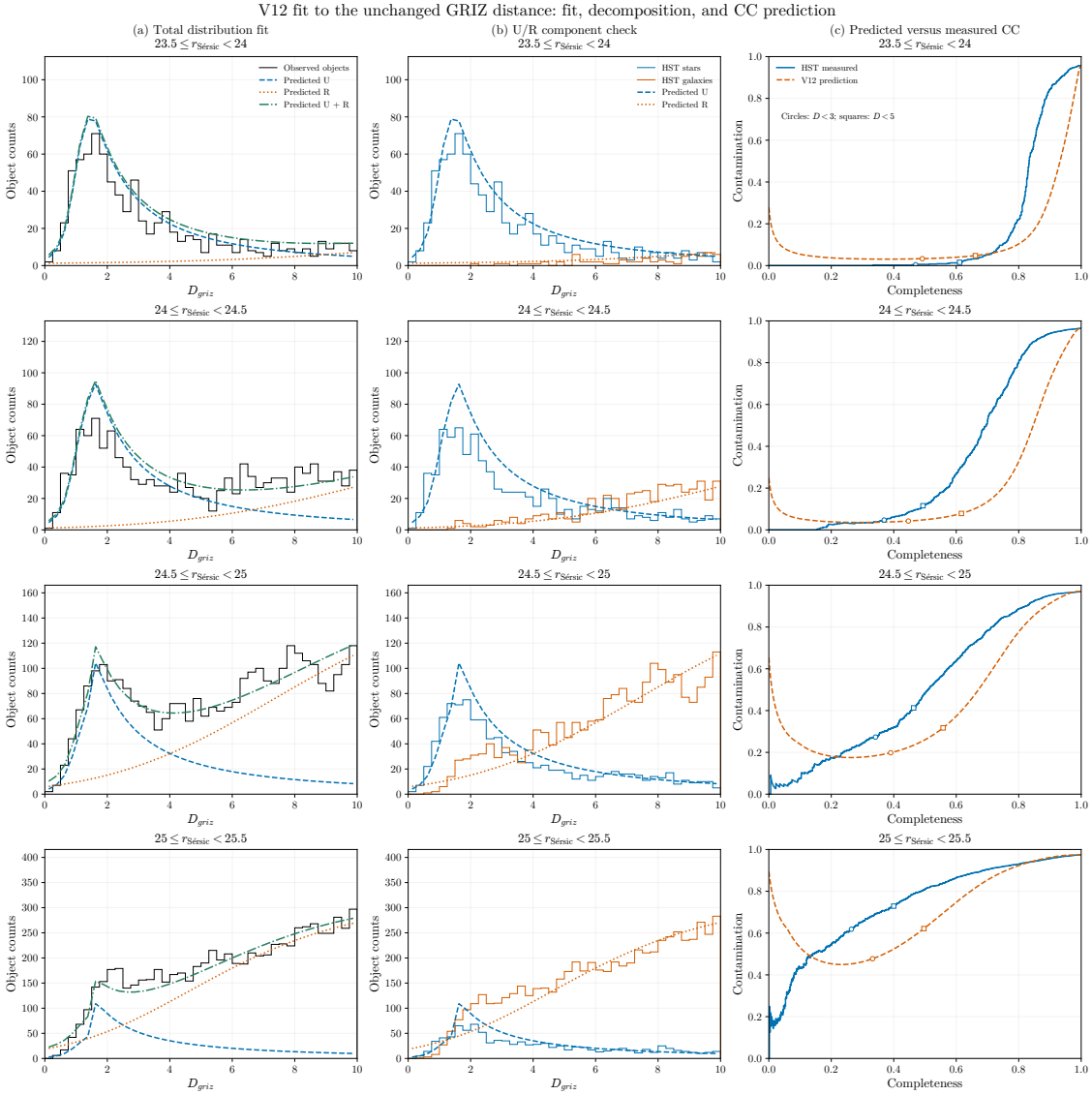


Figure 17. The panels in the first column are zoomed-in versions of Fig. 16, in four selected magnitude bins, as marked. Unresolved and resolved model components are shown with blue dashed and red dotted lines, respectively. The middle column is analogous to the left column, except that sources are separated by their HST labels (which were not used in model fitting). The right panels compare model-predicted and HST-observed contamination–completeness curves. The two symbols mark the locations of $D_{griz} = 3$ (circles) and $D_{griz} = 5$ thresholds.

Figure 18 compares the empirical performance of the *ri*, *riz* and *griz* band combinations to the *r*-band case, for a grid of D thresholds. As more bands are added, the sample contamination decreases at a given completeness level. For example, at a completeness level of 40% in the magnitude $24.5 \leq r < 25$, contamination decreases from 50% to 30% between the *r* and *griz* cases. At a completness level of 20%, the contamination decreases by more than a factor two. In practice, detailed optimization of the tradeoff between sample completeness and contamination depends on science case.

3.8. *Using D Metric-based Classification with Target Completeness or Contamination*

Given a two-component mixture model for a D distribution, we cannot not only predict class completeness and contamination but can choose one of them to be, for exampke, independent of magnitude. We focus on stellar class and D_{griz} because it gives the best performance.

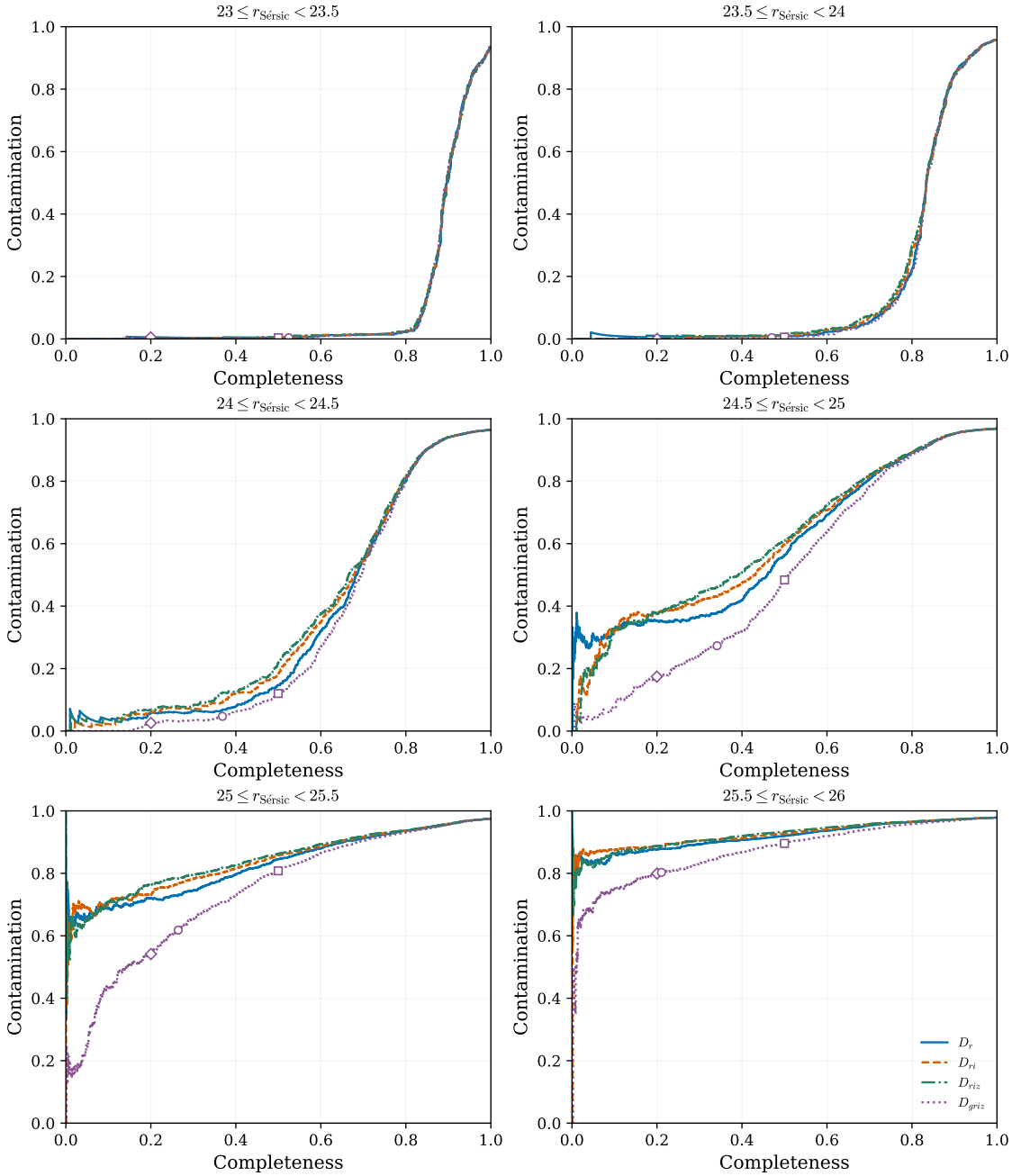


Figure 18. The performance of multi-band D classificators for stellar subsamples in different magnitude bins, as marked. Empirical contamination–completeness curves based on objects with HST classification labels are shown for the following band combinations: r, ri, riz and $griz$. All curves within a panel use the same cleaned HST reference sample with valid D in all required bands. Diamonds and squares mark approximately 20% and 50% stellar completeness on the $griz$ curves; circles mark $D_{griz} = 3$ threshold. Objects that are beyond the plotting window in Fig. 15 remain in the completeness and contamination denominators. Note that morphological information from additional bands improves performance.

Figure 19 shows variation of sample completeness and contamination measured using HST labels when model-based thresholds are selected to correspond to completeness levels of 20%, 50% and 90%, and contamination level of 20%. The agreement is remarkable. The values of D_{griz} as functions of magnitude required to achieve these target performance levels are shown in Figure 20. The values required for specified

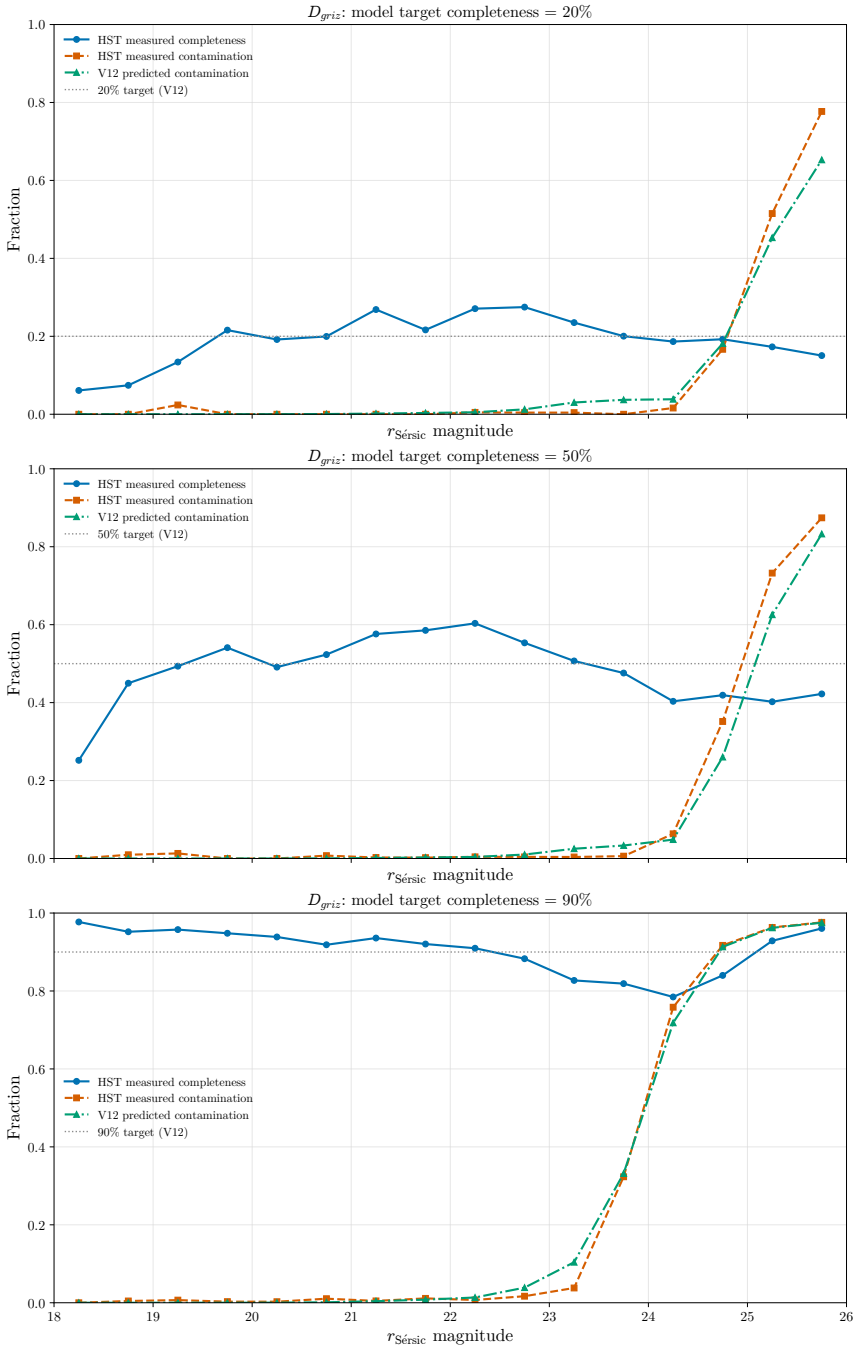


Figure 19. Target completeness application. We will fix caption after this figure is finalized. WE NEED TO ADD CURVES FOR MODEL-BASED CONTAMINATION. ALSO, WE NEED 4th PANEL THAT WOULD SHOW WHAT HAPPENS FOR TARGET CONTAMINATION OF 20%

completeness levels are expected to vary little over the sky. On the other hands, given that we analyzed the high-latitude COSMOS field, the implied contamination levels are an upper limit (closer to the Galactic plane the stellar counts will go up and thus the stellar sample contamination will go down).

3.9. *Comparison of Multiband D_{griz} Metric to* `griz_model_extendedness` *Parameter*

Rubin (R. Observatory Science Pipelines Developers et al. 2026) introduced a novel star-galaxy separation metric named `griz_model_extendedness`. The score is clipped to the interval [0, 1] but it is not a calibrated

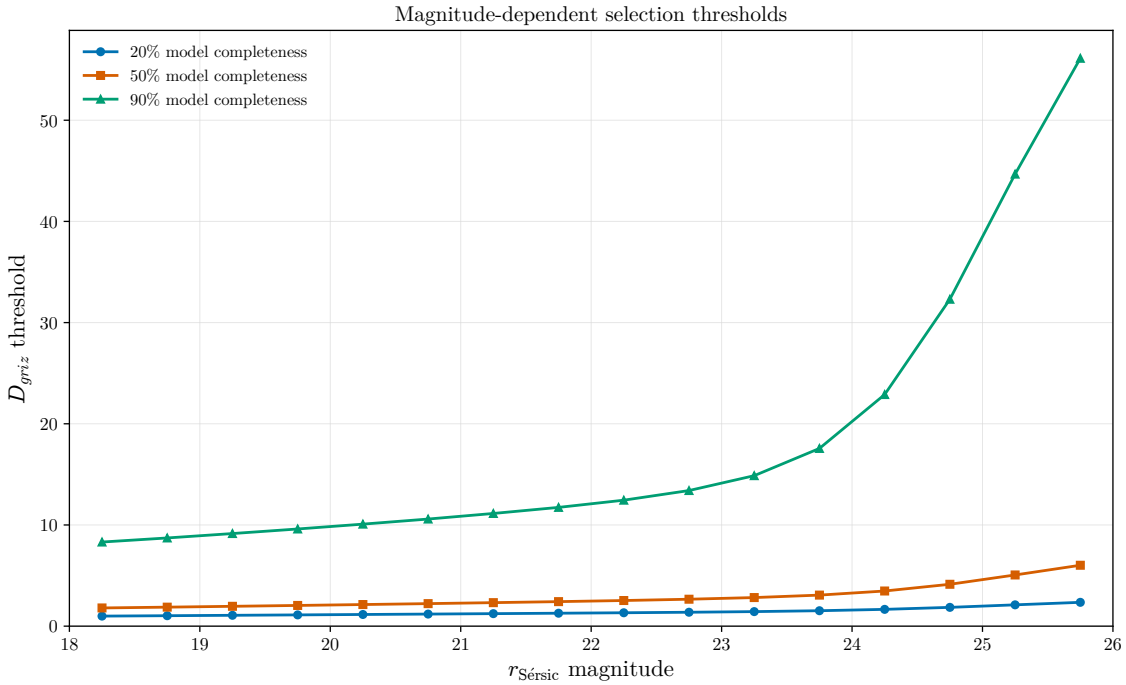


Figure 20. D values for target completeness application. We will fix caption after this figure is finalized. IT WOULD BE GOOD TO SHOW Y AXIS ON LOG SCALE.

probability that an object is a star or galaxy and should be be interpreted as a morphology ranking statistic. Given a subsets of bands, the algorithm separately adds PSF and Sérsic fluxes and then utilizes these total fluxes to define PSF a magnitude difference between PSF and Sérsic, as used here. In addition, the algorithm applies an empirical calibration and an uncertainty correction, and then modifies the result using a characteristic source size (for more details see Appendix B). The last two modifications do not have a theoretical underpinning. Adding fluxes from different bands is also not optimal (for example, in case of a barely resolved galaxy in red bands with a strong blue AGN component this addition could mask the fact that the source is a galaxy).

We tested `griz_model_extendedness` using HST labels and it shows a remarkable performance. Indeed, when trying to understand why its performance is so superior to Rubin's default `extendedness` parameter, we realized that the dominant reason is the use of Sérsic flux instead of CModel magnitudes. We compare the contamination–completeness curves for our D_{griz} classifier and `griz_model_extendedness` classifier in Figure 21. As evident, their performance is comparable, with `griz_model_extendedness` slighly better than D_{griz}. We show in Appendix B that we can reproduce `griz_model_extendedness` using cataloged parameters to apply uncertainty and source size corrections.

We reiterate that `griz_model_extendedness` is not a calibrated probability that an object is a star or galaxy, and by itself does not allow user to estimate completeness and contamination for a selected threshold. Figure 22 shows the variation of completeness and contamination for stellar selection using three different values of `griz_model_extendedness` threshold.

3.10. *Improved Stellar Classification Using Color Selection*

As shown in preceeding sections, the performance of morphological separation of unresolved and resolved sources starts to rapidly deteriorate about 2 magnitudes brighter than the 5σ limiting magnitude for point sources. Selection of unresolved objects is especially problematic because the sample is highly imbalanced at the faint end. At the same time, samples of faint stars are valuable for studying the Milky Way structure (e.g. stellar streams in the halo). Here we investigate if color selection can be used to decrease contamination without significant decrease in completeness of such samples.

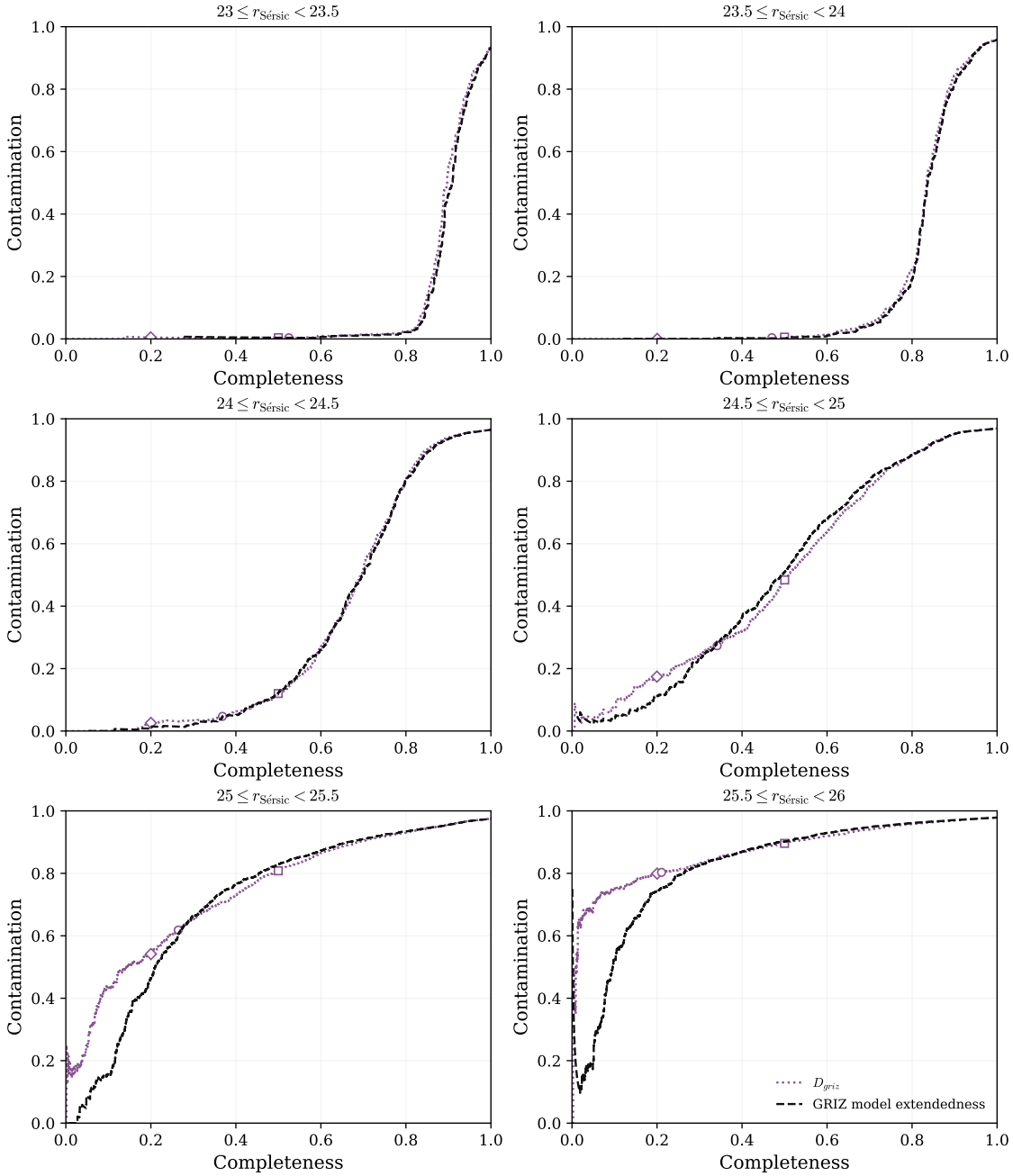


Figure 21. A comparison of completeness–contamination curves for two different classification methods based on D_{griz} developed here and `griz_model_extendedness` computed by Rubin Science Pipelines.

Figure 23 compares color distributions of morphologically selected samples analyzed in preceeding sections. As evident, HST galaxies (misclassified as stars) appear systematically bluer than stars, except for the $z-y$ color. Therefore, selection based on bluer colors can result in improved classification.

Motivated by color distributions shown in Figure 23, we investigate simple $g-r$ and $r-i$ selection criteria in Figure 24. The $g-r>0.3$ cut was chosen as appropriate for selecting blue stars near the turn-off point on the principal stellar locus (composed of main sequence and red giant stars; see e.g., L. Palaversa et al. 2026). The cut $g-r>1.0$ selects red (M type) stars and excludes almost all galaxies. The cut $r-i>0.8$ is similar but more effective at excluding galaxies.

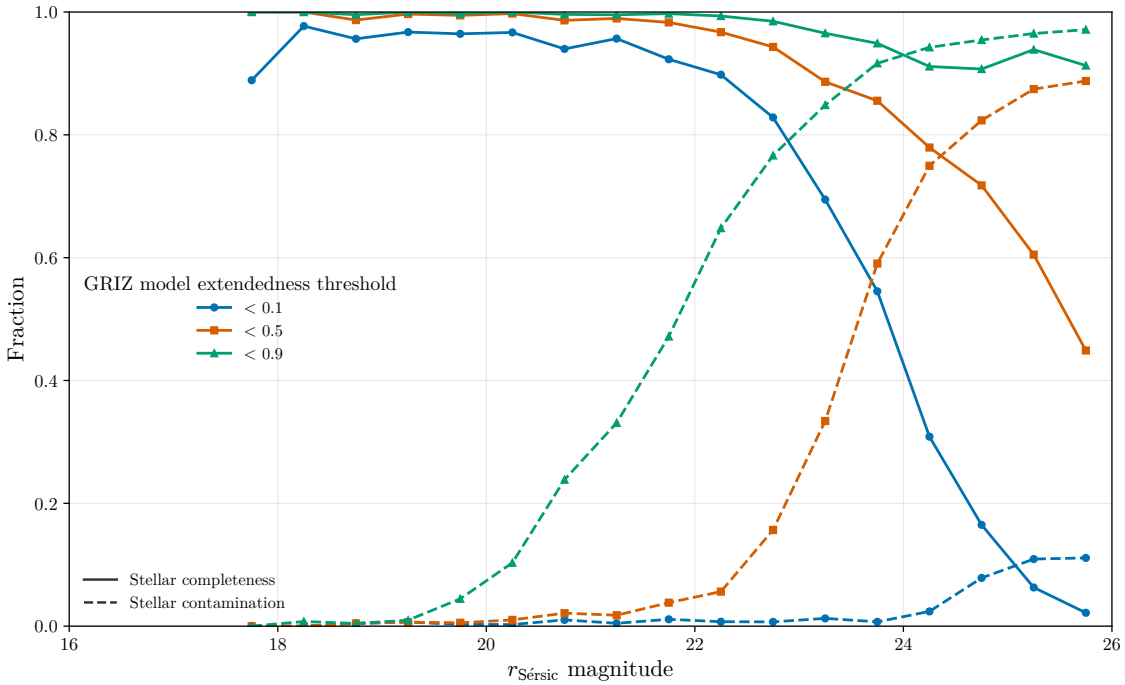


Figure 22. Performance vs. magnitude for a fixed threshold. IF we keep this figure, we will fix caption after this figure is finalized.

The top panel in Figure 24 shows substantial decrease of unresolved sample contamination when $g-r > 0.3$ condition is added, both in single-band and multiband cases. For example, for $24 \leq r < 24.5$ bin at fiducial stellar completeness of 50%, D_{griz} and $g - r > 0.3$ selection decrease contamination to 25% from 45% in the r band case without a color cut; at stellar completeness of 20%, the contamination level is even more dramatically decreased from 40% to 12%. Corresponding values for $24.5 \leq r < 25$ are 78% to 63%, and 72% to 54%.

Decrease of contamination is much more substantial in case of red stars. For example, even in the fainter bin, $24.5 \leq r < 25$, the addition of $g - r > 1.0$ criterion results in contamination as low as 20% with substantial completeness of 45%. When completeness of 20% is acceptable (relative to all stars, not just the red subsample), contamination levels as low as 10% can be achieved.

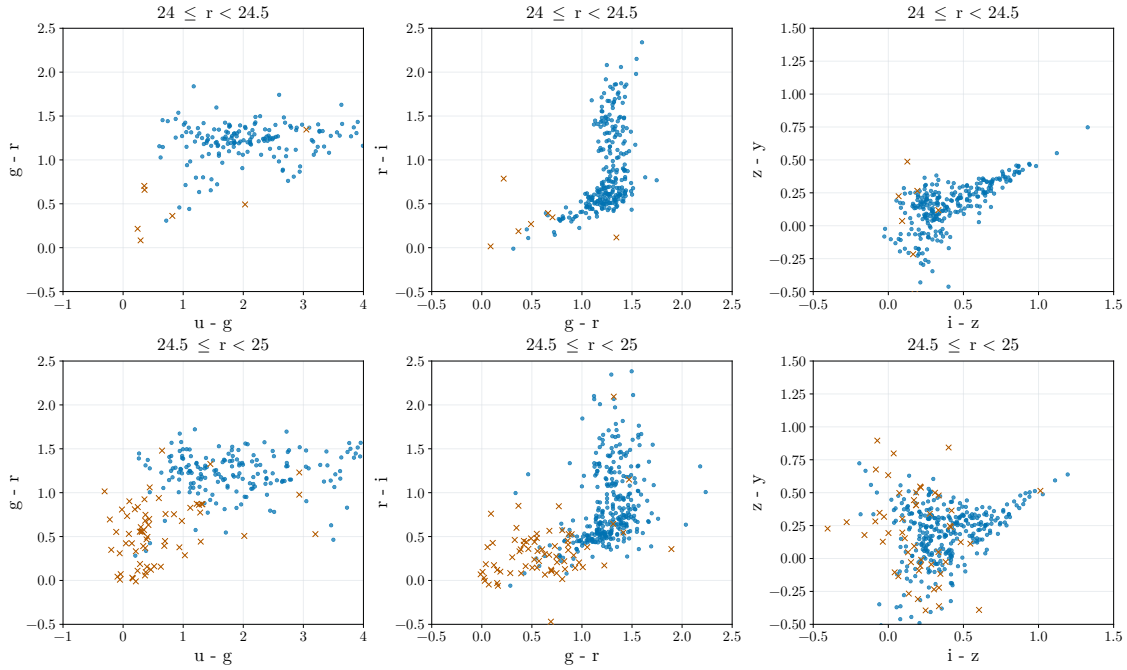


Figure 23. The color distribution of objects morphologically classified as unresolved using Rubin DP2 measurements (with D_{griz} thresholds selected to ensure 20% completeness), and then separated into stars (blue dots) and galaxies (red crosses) using HST labels, in two magnitude bins: $24 \leq r < 24.5$ (top) and $24.5 \leq r < 25$ (bottom). The fraction of HST-classified galaxies in these two subsamples is 2.6% and 18%, respectively. Note that galaxies appear systematically bluer than stars, except for the $z - y$ color, which allows for improved color-based classification.

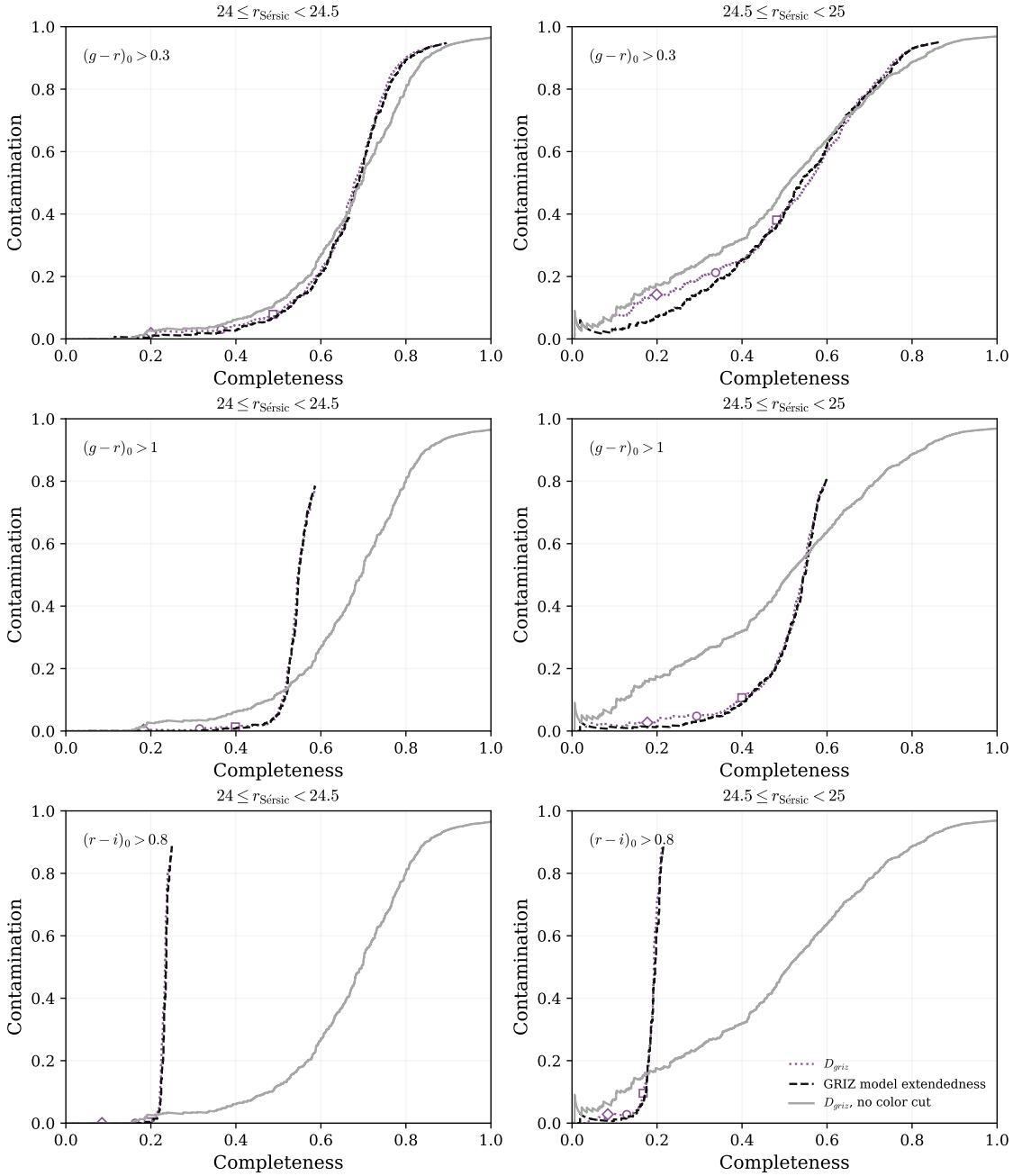


Figure 24. The performance of color-aided multi-band D classificators for stellar subsamples in two magnitude bins: $24 \leq r < 24.5$ (left) and $24.5 \leq r < 25$ (right), and for three color cuts: $g-r > 0.3$ (top), $g-r > 1.0$ (middle) and $r-i > 0.8$ (bottom). In each panel, empirical contamination-completeness curves based on objects with HST classification labels are shown for the following band combinations: r, ri, riz and $griz$. A curve corresponding to the $griz$ case without color cut is also shown for reference. Note the marked decrease of sample contamination when morphological classification is aided with color selection. Diamonds and squares mark the $griz$ thresholds corresponding to approximately 20% and 50% completeness before the color cut, respectively; circles mark $D_{griz} < 3$. Open gray diamonds show the 20% completeness selection without a color cut.

APPENDIX

A. Iterative fitting procedure

The data distribution in the $x \equiv \Delta m$ vs. $m \equiv m_{\rm Sersic}$ plane is fit with a model given by eq. 3. The basic idea of the iterative fitting algorithm is to constrain $f_{\rm unr}(x)$ and $f_{\rm res}(x)$ templates first (see eqs. 7–9 and eq. 12), and then constrain their relative normalization as a function of magnitude, while allowing for template shifts and constrained changes of their initially estimated shapes.

We first estimate $f_{\rm unr}(x)$ by selecting input data with $20 < m < 22.5$ and $|x| < 0.05$ and fit a model with 7 free parameters: f, μ_c, σ_c, γ, δ, h, and k. In the final fitting step, we keep γ, δ, h, and k fixed to their best-fit values and allow the location and scale of $f_{\rm unr}(x)$, μ_c and σ_c, and its tail-to-core normalization, f, to vary by fitting them as smooth functions of magnitude (see below).

We estimate the shape of $f_{\rm res}(x)$ using a subset of data with $23 < m < 24$ and $0.05 < x < 1.5$. We model it with $K = 2$ skew-normal distributions (see eq. 12) and obtain 7 best-fit parameters (the sum of the two w_R parameters must be unity by definition). In the final fitting step, we keep α_R parameters fixed to their best-fit values and allow w_R, μ_R and σ_R to vary as smooth functions of magnitude.

In the final step, we fit full two-dimensional data distribution $f(x, m)$,

$$f(x,m) = w_U(m)\,[1 - f(m)]\,J(x,m) + w_U(m)\,f(m)\,T(x,m) + [1 - w_U(m)]\,R(x,m), \quad (1)$$

where $R(x, m)$ corresponds to eq. 12, using maximum likelihood method. We minimize function

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N} \ln(p_i), \quad (2)$$

where $p_i = f(x_i, m_i)$ and x_i and m_i are data for N objects.

The key aspect of this final step is to fit normalization functions $w_U(m)$ and $f(m)$ as smooth functions of magnitude, as described by eqs. 10 and 11, respectively. We allow for the shift of the location and scale of each template component in eq. 1 (J, T, and R) with magnitude, as well as for their shape changes, constrained by keeping a subset of model parameters fixed to their best-fit values determined in the initial fitting steps described above.

We assume a linear variation of μ_c with magnitude, and for σ_c adopt a photometric error model from B. Sesar et al. (2007) (their eq. 5),

$$\sigma_c(m) = \sqrt{\sigma_0^2 + a^2 \cdot 10^{\,b\,(m-22)}}, \quad (3)$$

where σ_0, a, and b are free model parameters. For w_R, μ_R and σ_R, free parameters that describe $R(x, m)$, we assume a linear variation with magnitude. This final fitting step includes 19 free parameters. With the five parameters determined in the initial steps, our full 2D model includes 24 free parameters.

XXX: we might keep R fixed to its best-fit for HST galaxies, we'll see how it goes...

B. Reproducing `griz_model_extendedness` parameter

Here Daniel describes corrections for uncertainties and size, and shows a plot that compares CC curves for our basic D_{griz}, corrected D_{griz}, and for `griz_model_extendedness`.

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